GFE 306

RENEWABLE ENERGY SYSTEMS AND TECHNOLOGY

COMPREHENSIVE UNDERGRADUATE LECTURE MATERIAL

Expanded explanations • worked engineering examples • redrawn schematic diagrams • Nigerian applications • tutorial problems

Course load: 3 Units | 45 Lecture Hours

Edition: LUMINA Comprehensive Edition — expanded teaching draft

Academic-use note. This edition is an expanded teaching resource based on the supplied 67-page GEE/GFE 306 lecture note. The supplied source labels the course “GEE 306” on its opening page, while this LUMINA edition uses “GFE 306” to match the course label used in the current LUMINA workflow. Confirm the official departmental code against the institution’s course catalogue before formal submission.

How to use this material

This material is organised around the 15 units in the supplied lecture note. Each unit develops the source topics into a teachable sequence: concepts and definitions, physical principles, system components, operating steps, applications, limitations, engineering checks and examples. The supplementary topics in the source—energy comparisons, end-use systems, hybrid systems, planning, Nigerian policy context, smart grids, environmental impacts and life-cycle thinking—are retained after Unit 15.

A diagram in this edition is an original educational schematic redrawn for legibility. It is not a construction drawing or a substitute for manufacturer drawings, standards, site measurements or detailed engineering design.

Course description

Renewable Energy Systems and Technology examines conventional and renewable energy resources, extraction and conversion processes, energy storage and transmission, conservation, energy economics, environmental impacts and sustainable development. The course considers fossil fuels, nuclear energy, solar, biomass, wind, hydropower and geothermal energy, with emphasis on their roles in Nigeria and the wider energy system.

Learning outcomes

- Define energy, power and renewable energy and explain their significance.
- Classify primary, secondary, final and useful energy.
- Explain the stages of energy conversion and the operation of major energy technologies.
- Analyse solar, biomass, wind, hydro and geothermal systems.
- Perform basic engineering calculations for energy, power, efficiency, storage, transmission and economics.
- Compare technologies using technical, economic, environmental and social criteria.
- Explain energy security, energy planning, smart grids and sustainable development.

Course organisation

Unit	Topic	Hours
1	Energy concepts and energy systems	3
2	Global and Nigerian energy systems	3
3	Energy resources and extraction	3
4	Fossil-fuel energy systems	3
5	Nuclear energy systems	3
6	Solar energy	3
7	Biomass energy	3
8	Wind energy	3
9	Hydropower	3
10	Geothermal and other renewable energy	3
11	Energy conversion and power systems	3
12	Energy storage	3
13	Energy transmission, conservation and efficiency	3
14	Energy economics and energy mix	3
15	Sustainable energy and sustainable development	3
Total	15 units	45

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08. 8. Wind energy
09. 9. Hydropower
10. 10. Geothermal and other renewable energy
11. 11. Energy conversion systems
12. 12. Energy storage
13. 13. Transmission, conservation and efficiency
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15. 15. Sustainable energy and development
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19. Selected technical references

UNIT 1 — INTRODUCTION TO ENERGY SYSTEMS

Unit purpose. Establish the language and basic physical quantities used throughout energy engineering. A student should be able to distinguish energy from power, identify common energy forms, classify energy by its position in the supply chain and trace energy through a conversion system.

1.1 Meaning and measurement of energy

Energy is the capacity of a system to perform work or produce a physical change. In engineering, energy is a measurable property that may be stored, transferred or converted. It is not destroyed in an ideal accounting of the complete system; rather, it changes form or is transferred to the surroundings. The SI unit is the joule (J).

Unit	Meaning / relationship
1 kJ	1,000 J
1 MJ	1,000,000 J
1 Wh	3,600 J
1 kWh	3.6 MJ = 3,600,000 J
1 MWh	1,000 kWh
Btu	British thermal unit; commonly used in fuel and heating calculations
toe	Tonne of oil equivalent; an energy-accounting unit

Electrical energy is often billed in kilowatt-hours. A kilowatt-hour is an amount of energy, not a rate of energy use. A 1 kW appliance operating for one hour consumes 1 kWh, assuming it operates at its rated power throughout that hour.

1.2 Power and the energy–time relationship

Power is the rate at which energy is transferred or converted. It describes how quickly a device uses or delivers energy. The SI unit is the watt (W), equivalent to one joule per second.

$$P = E / t \text{ therefore } E = P\,t$$

Where P is power in watts, E is energy in joules and t is time in seconds. For electrical energy expressed in kWh, multiply power in kW by operating time in hours.

Worked Example 1.1 — energy consumed by a heater

Problem. A 2 kW electric heater operates continuously for 5 hours. Determine the electrical energy consumed in kWh and MJ.

Solution.

Step 1. Use E = P × t.

Step 2. Substitute P = 2 kW and t = 5 h: E = 2 × 5 = 10 kWh.

Step 3. Convert to megajoules using 1 kWh = 3.6 MJ: E = 10 × 3.6 = 36 MJ.

Final answer: 10 kWh, equivalent to 36 MJ.
Engineering meaning: The heater’s power rating is 2 kW; the total energy depends on how long it operates.

1.3 Forms of energy

Mechanical energy

Mechanical energy is associated with motion and position. Kinetic energy is the energy of motion; gravitational potential energy is associated with elevation in a gravitational field.

$$\text{Kinetic energy: } E_k = \tfrac{1}{2} m v^2 \text{ | Gravitational potential energy: } E_p = m g h$$

Here m is mass (kg), v is speed (m/s), g is gravitational acceleration (approximately 9.81 m/s² near Earth's surface) and h is height above a chosen reference level (m). These relationships explain why elevated water stores potential energy and why moving air and water possess kinetic energy.

Thermal energy

Thermal energy is associated with the microscopic motion and interactions of particles in matter. A temperature difference can drive heat transfer by conduction, convection or radiation. In thermal power stations, fuel energy is converted to heat, which is then used to produce steam or hot gas.

Chemical energy

Chemical energy is associated with the arrangement of atoms and electrons in chemical substances. Combustion releases part of this stored energy. Petrol, natural gas, coal and biomass are chemical energy carriers. Batteries also store energy electrochemically; during discharge, chemical reactions drive an electric current.

Electrical, nuclear and electromagnetic energy

Electrical energy is associated with electric charges and electric fields. Nuclear energy is associated with the binding energy of atomic nuclei and can be released through fission or fusion. Electromagnetic or radiant energy is carried by electromagnetic waves; sunlight is the key input for solar technologies.

1.4 Primary, secondary, final and useful energy

These terms describe different points in the energy supply chain. Primary energy is obtained directly from nature before conversion. Secondary energy is produced by converting primary energy. Final energy is delivered to the user. Useful energy is the portion that actually provides the desired service.

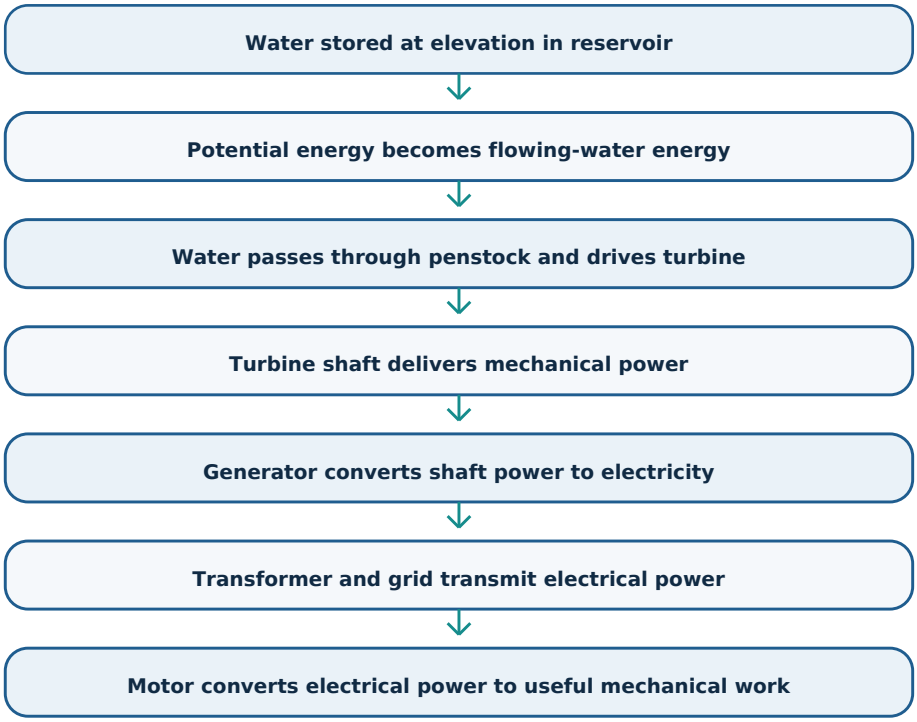
Category	Definition	Examples
Primary	Energy in natural resources before conversion	Crude oil, coal, sunlight, wind, flowing water, uranium, biomass
Secondary	Energy carrier produced by conversion	Electricity, hydrogen, refined petrol and diesel
Final	Energy delivered to an end user	Electricity at a factory meter, LPG cylinder, petrol in a vehicle
Useful	Energy output that provides the service	Light from a lamp, shaft work from a motor, heat in a cooking pot

A useful way to remember the difference is to follow crude oil: crude oil is primary energy; refined petrol is secondary energy; petrol delivered to a vehicle is final energy; the mechanical work delivered by the engine is useful output. Losses occur during extraction, refining, transport and end-use conversion.

1.5 General energy conversion chain

An energy system links a resource to an energy service. A typical chain includes resource collection, conversion, conditioning, storage or transmission, distribution and end use. Each stage has a purpose and may introduce losses.

Hydroelectric energy conversion chain



For a hydroelectric plant, the reservoir provides hydraulic head. Water flows through the intake and penstock, transferring energy to the turbine runner. The turbine rotates a generator. A transformer raises the voltage for transmission; at the receiving end, substations reduce voltage for distribution and use.

1.6 Energy losses and efficiency

No real conversion device converts all input energy into the desired output. Some energy leaves as heat, sound, vibration, friction, exhaust or electrical resistance. Efficiency is the ratio of useful output to input, expressed as a fraction or percentage.

η = (Useful energy output / Energy input) × 100%

Do not compare quantities with incompatible units. Convert both input and output to the same energy unit before calculating efficiency.

Unit 1 — check your understanding

- Explain why kWh is a unit of energy even though it contains the word “hour”.
- Distinguish primary energy from final energy using a solar home system as an example.
- A 750 W pump operates for 4 hours. Calculate its energy use in kWh.
- Explain two conversion losses in a hydroelectric system.

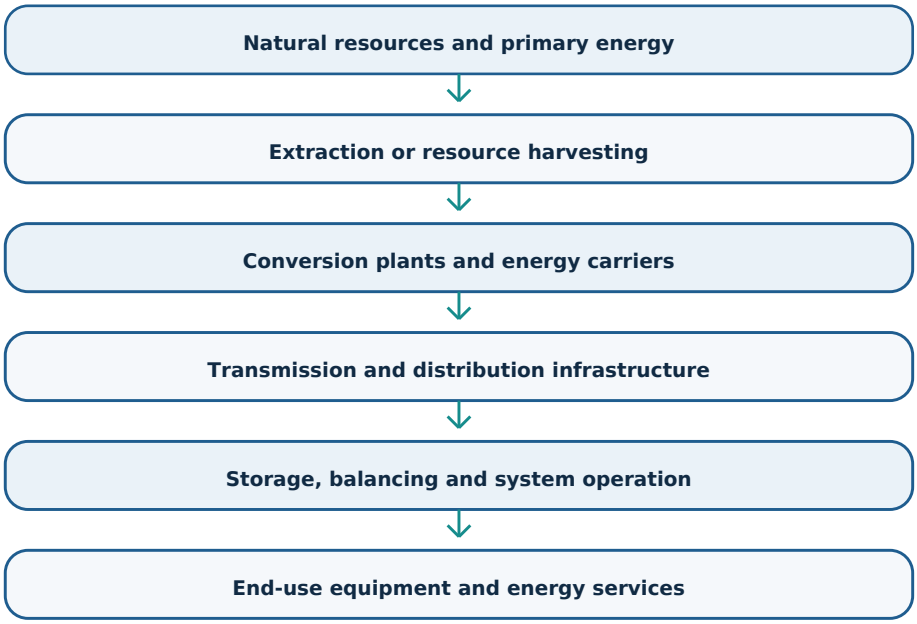
UNIT 2 — GLOBAL AND NIGERIAN ENERGY SYSTEMS

Unit purpose. Understand energy as an interconnected system rather than a collection of power plants. The unit also places Nigeria’s energy resources, infrastructure and transition priorities in context.

2.1 Structure of the global energy system

The global energy system links resources to energy services through a chain of physical infrastructure, institutions and consumers. Its principal stages are resource assessment, extraction or harvesting, conversion, transmission, distribution, storage and end use. A failure or bottleneck at any stage can limit the service delivered even where the resource itself is abundant.

Main elements of an energy system



2.2 Major energy-consuming sectors

Sector	Typical energy services	Examples of equipment
Electricity supply	Generation and grid support	Turbines, generators, transformers, batteries
Transport	Movement of people and goods	Petrol/diesel engines, electric drivetrains, rail
Industry	Process heat, motion and materials processing	Motors, furnaces, boilers, compressors
Residential	Cooking, lighting, cooling, refrigeration	Lamps, cookers, fans, refrigerators
Commercial	Lighting, cooling, computing and services	HVAC, lifts, servers, pumps
Agriculture	Irrigation, drying, processing and cold chains	Pumps, dryers, tractors, chillers

Energy demand differs by sector. Industrial demand may be dominated by motors and process heat, while household demand may be concentrated in cooking, lighting, cooling and appliances. Energy planning must therefore start with the service required, not merely with a preferred generating technology.

2.3 Global energy transition

The energy transition is the gradual structural change in how energy is produced, transported and used. It includes increasing renewable generation, improving efficiency, electrifying end uses where practical, modernising grids, deploying storage and reducing emissions from existing systems. The supplied lecture note identifies falling technology costs, energy security, climate policy, electrification, energy-access programmes and private investment as important drivers.

Installed capacity and actual energy generation are different measures. Capacity (MW or GW) describes the maximum rated power at a moment; generation (MWh or TWh) measures energy produced over time. A technology with a high installed capacity may generate less annual energy if its resource is intermittent or its capacity factor is low.

2.4 Nigeria’s energy situation

The supplied lecture note identifies crude oil, natural gas and coal among Nigeria’s fossil resources, and solar, hydropower, biomass, wind and possible geothermal resources among renewable opportunities. The central challenge described is the gap between substantial natural-resource endowment and the practical delivery of reliable, affordable energy services. Infrastructure, transmission constraints, distribution performance, investment, rural access and affordability all influence outcomes.

2.5 Nigeria Energy Transition Plan

The supplied source describes Nigeria’s Energy Transition Plan as a pathway targeting net-zero emissions by 2060 while addressing energy poverty and economic development. It identifies five focus sectors: power, cooking, transportation, industry, and oil and gas. The source also describes natural gas as a transition fuel in the plan while renewable technologies expand. Policy targets and implementation details can change; students should consult the current official plan and subsequent government updates when using numerical targets in project work.

2.6 Renewable-energy opportunities across Nigeria

Resource	Opportunity identified in the supplied lecture note	Engineering considerations
Solar	Nationwide potential, particularly northern Nigeria	Irradiance, shading, orientation, storage, dust, temperature and grid access
Hydropower	Rivers and sites with suitable head	Flow-duration data, head, sediment, ecology, seasonal variation
Biomass	Agricultural and forestry residues	Feedstock quantity, moisture, collection radius, competing uses and sustainability
Wind	Selected northern and coastal locations	Measured wind-speed distribution, hub height, turbulence and grid connection
Geothermal	Potential requiring further exploration	Subsurface temperature, geology, drilling risk and resource confirmation
Waste-to-energy	Major cities and industrial centres	Waste composition, sorting, moisture, emissions control and residual disposal

2.7 Reliability, access and affordability

A technically sound energy system must deliver power when and where it is needed. Reliability concerns continuity and quality of supply. Access concerns whether households, businesses and public services can connect to and use modern energy. Affordability concerns both initial connection costs and recurring bills. A system can perform well on one dimension and poorly on another; for example, a remote solar home system may improve access but have limited power for high-demand appliances.

Unit 2 — applied discussion

- Why can a country with substantial fuel resources still experience unreliable electricity?
- Explain the difference between installed capacity and annual electricity generation.
- Choose a suitable energy technology for a remote clinic and justify the choice using demand, resource and maintenance considerations.

UNIT 3 — ENERGY RESOURCES AND EXTRACTION

Unit purpose. Classify resources and understand how primary energy is obtained before it enters conversion and supply networks.

3.1 Renewable and non-renewable resources

A renewable resource is replenished by natural processes on a timescale relevant to human use, provided extraction or harvesting does not exceed regeneration. Solar radiation, wind, flowing water and geothermal heat are commonly treated as renewable. Biomass is renewable only when the biological feedstock is replenished and managed sustainably. A non-renewable resource is finite on human timescales; examples include coal, crude oil, natural gas and uranium.

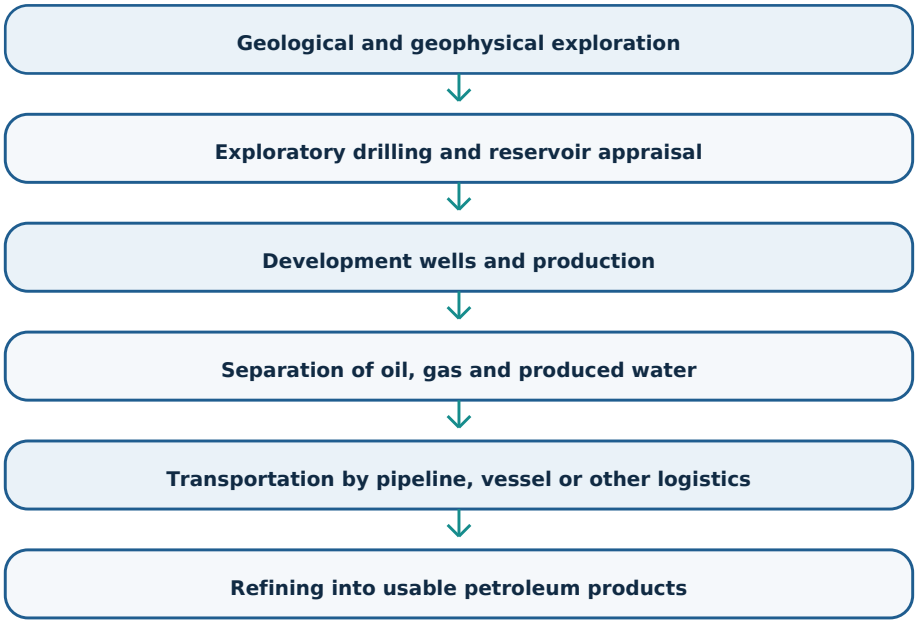
Characteristic	Renewable resources	Non-renewable resources
Availability	Naturally replenished, subject to local limits	Finite stock depleted by extraction
Fuel input	Often no purchased fuel for solar/wind/hydro; biomass requires feedstock	Fuel extraction, processing and supply required
Variability	Some resources vary with weather, season or time	Many plants are dispatchable if fuel and plant are available
Environmental effects	Often lower operational emissions; not impact-free	Fossil combustion and extraction can cause substantial impacts
System needs	May require storage, flexible generation and grid upgrades	Fuel logistics, pollution controls and waste management

3.2 Resource assessment before extraction

Resource assessment estimates the quantity, quality, location and temporal behaviour of an energy resource. Engineers use geological surveys and exploratory wells for fossil resources; solar radiation measurements and satellite data for solar; wind measurements at relevant heights for wind; flow and head measurements for hydro; and feedstock surveys for biomass. A resource estimate is not the same as a bankable project assessment: the latter also evaluates land, access, grid connection, environmental constraints, permits, cost and risk.

3.3 Oil extraction and processing chain

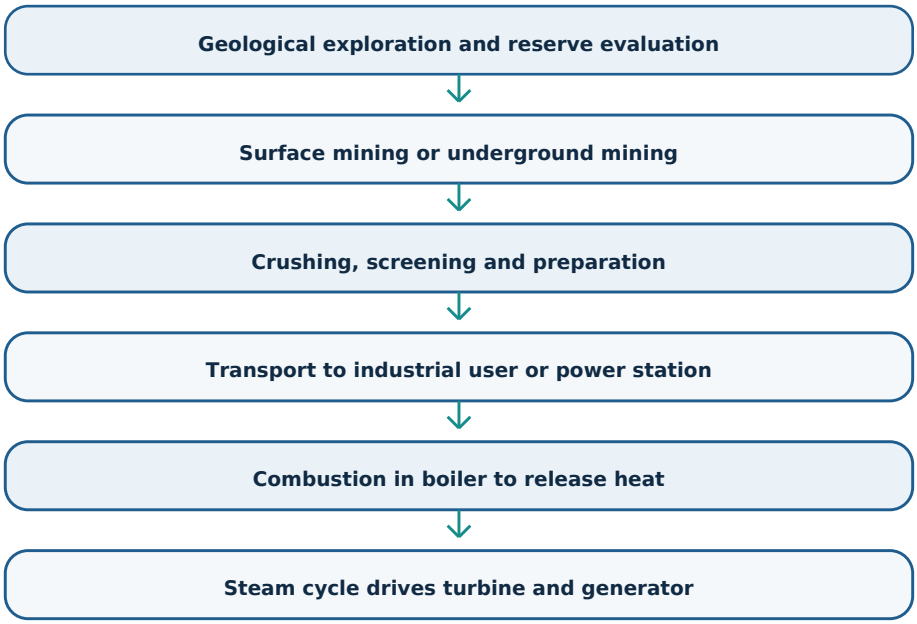
Typical petroleum supply chain



Exploration identifies possible hydrocarbon-bearing formations. Drilling provides access to the reservoir. During production, pressure and artificial-lift systems may bring fluids to the surface. Produced fluids are separated, treated and transported. Refining then separates and converts crude oil into products such as petrol, diesel, kerosene and other fractions. Each stage has technical, safety and environmental requirements.

3.4 Coal extraction chain

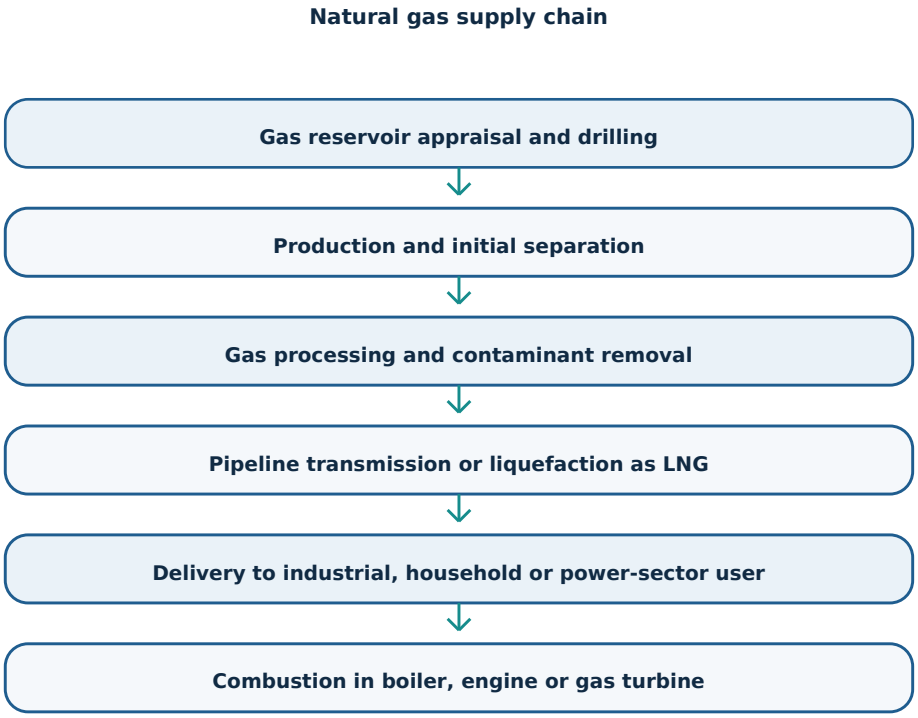
Coal supply and conversion chain



Coal rank is commonly described from lower to higher carbon content and heating value as lignite, sub-bituminous, bituminous and anthracite. Actual properties vary with deposit. Mining method depends on

seam depth, geology, overburden and safety. Coal preparation can remove some mineral impurities and improve consistency, but combustion still requires emissions management and ash handling.

3.5 Natural gas extraction and delivery



Natural gas consists mainly of methane, although its composition varies. Processing may remove water, carbon dioxide, hydrogen sulphide and heavier hydrocarbons depending on the gas and its intended use. Pipeline transport uses compression to maintain flow. Liquefied natural gas (LNG) is cooled to cryogenic temperature to reduce volume for shipping; it must be regasified before ordinary pipeline use.

3.6 Extraction impacts and responsible practice

Resource extraction can affect land, water, air, biodiversity and communities. Relevant controls include environmental impact assessment, safe well design, mine ventilation and ground support, spill prevention, produced-water treatment, methane leak detection, rehabilitation and closure planning. An energy engineer should consider the full resource pathway rather than treating fuel as impact-free once it reaches a power station.

Unit 3 — application

A proposed biomass plant near a rice-growing area should not assume all rice husks are available. Some residues may already be used as animal bedding, soil amendment, fuel or industrial input. A credible feedstock assessment measures seasonal quantities, moisture, collection cost, competing uses and sustainable removal rates.

UNIT 4 — FOSSIL-FUEL ENERGY SYSTEMS

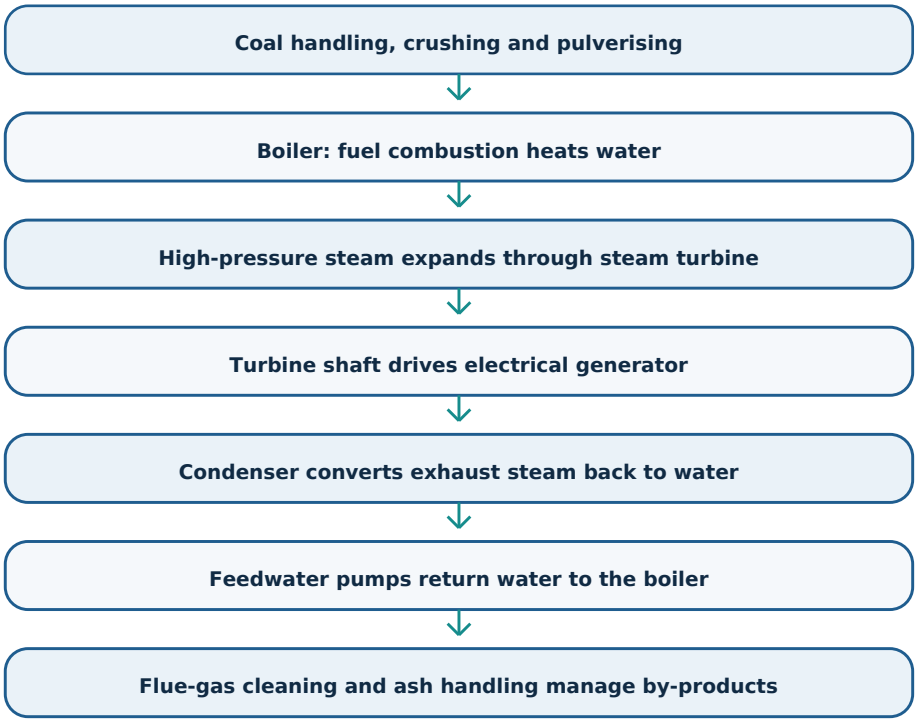
Unit purpose. Explain how coal and natural gas are converted into electricity, identify major plant components and evaluate the efficiency and environmental trade-offs of fossil-fuel generation.

4.1 Coal and coal properties

Coal is a combustible sedimentary rock formed from ancient organic matter subjected to burial, pressure and heat over geological time. Its grade influences moisture, volatile matter, fixed carbon, ash content and heating value. These properties affect boiler design, combustion behaviour, emissions and the amount of fuel needed per unit of electricity.

4.2 Coal-fired steam power station

Coal-fired steam power plant — simplified process



The boiler transfers heat from combustion gases to water and steam. High-pressure steam expands through turbine stages and produces shaft work. The generator converts shaft work into electrical energy. The condenser rejects low-grade heat and returns steam to liquid water, allowing the cycle to repeat. Pumps restore pressure in the feedwater circuit. The plant also requires fuel conveyors, pulverisers, fans, cooling systems, emissions controls and ash handling.

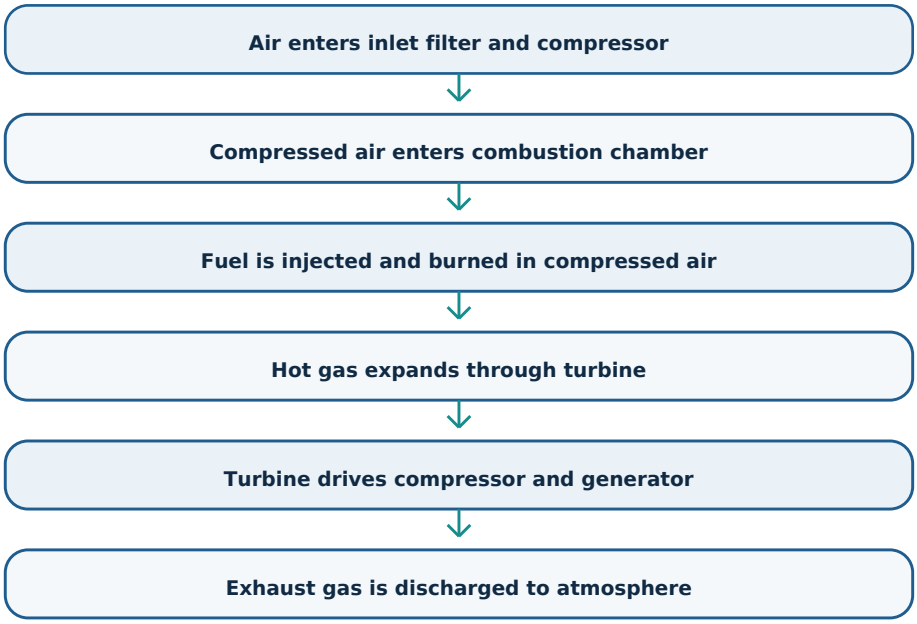
4.3 Natural gas and combustion

Natural gas is composed mainly of methane (CH_4). Complete combustion with sufficient oxygen produces carbon dioxide, water and heat. Incomplete combustion can produce carbon monoxide and unburned hydrocarbons. Actual combustion systems must control air–fuel ratio, flame stability, temperature and emissions.

$$CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O + \text{heat}$$

4.4 Open-cycle gas turbine

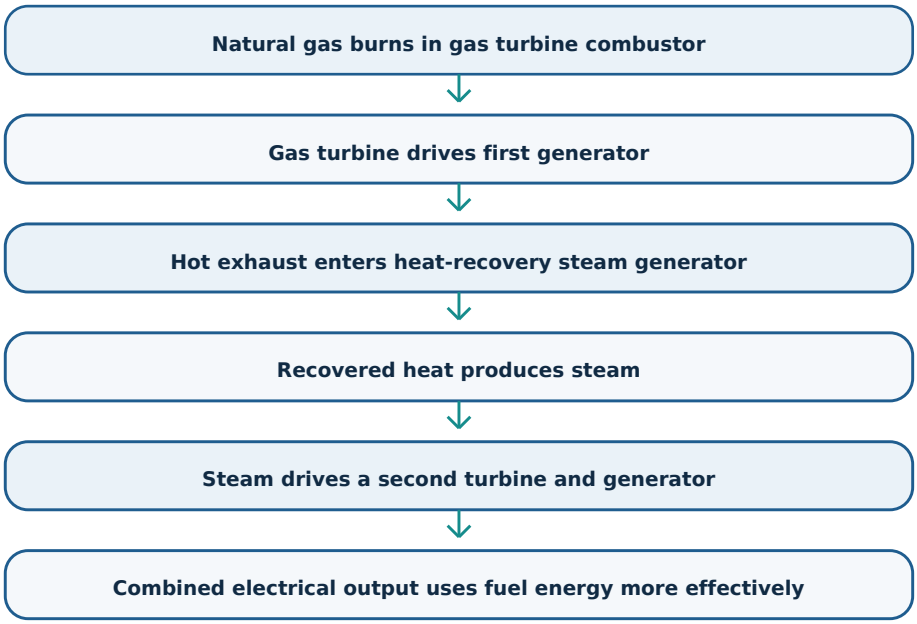
Open-cycle gas turbine plant



A gas turbine follows the Brayton cycle. The compressor raises air pressure; fuel is burned in the combustor; hot gases expand through the turbine. Part of turbine output drives the compressor, and the remaining shaft power drives the generator. Gas turbines can start relatively quickly compared with many steam plants, but their efficiency depends on design, load, ambient temperature and operating conditions.

4.5 Combined-cycle gas turbine (CCGT)

Combined-cycle gas turbine arrangement



A combined-cycle plant recovers heat from the gas-turbine exhaust to produce steam for a steam turbine. It therefore obtains useful work from both the gas-turbine cycle and a secondary steam cycle. The improved

efficiency comes from using exhaust heat that would otherwise be rejected. The system requires a heat-recovery steam generator, steam turbine, condenser, feedwater system and additional controls.

4.6 Advantages, limitations and impacts

- Advantages: high energy density, established supply chains, dispatchable generation when fuel is available, and suitability for large-scale output.
- Limitations: finite fuel, fuel-price exposure, greenhouse-gas emissions, air pollutants, extraction impacts, water use and waste/by-product management.
- Coal plants require management of particulate matter, sulphur oxides, nitrogen oxides, carbon dioxide and ash. Gas plants generally emit less carbon dioxide per unit of electricity than coal plants, but methane leakage across the gas supply chain can materially affect climate impact.

Worked Example 4.1 — thermal power-plant efficiency

Problem. A thermal power station receives 1,000 MJ of chemical energy from fuel and delivers 350 MJ of electrical energy. Calculate its gross conversion efficiency.

Solution.

Step 1. Use η = (electrical output energy / fuel input energy) × 100%.

Step 2. Substitute η = (350 MJ / 1,000 MJ) × 100%.

Step 3. Evaluate the ratio: 0.35 × 100% = 35%.

Final answer: 35%.
Engineering meaning: The remaining 65% is not all one type of loss; it is rejected or consumed through exhaust heat, cooling, mechanical and electrical losses, and auxiliary equipment.

Unit 4 — design questions

- Why does a combined-cycle plant achieve higher efficiency than a simple gas turbine under suitable conditions?
- List the major energy flows in a coal-fired steam cycle.
- Explain why fuel quality and ambient temperature can affect plant output.

UNIT 5 — NUCLEAR ENERGY SYSTEMS

Unit purpose. Introduce nuclear fission and fusion, the controlled chain reaction, the basic nuclear electricity-generation pathway and the engineering responsibilities associated with safety and waste.

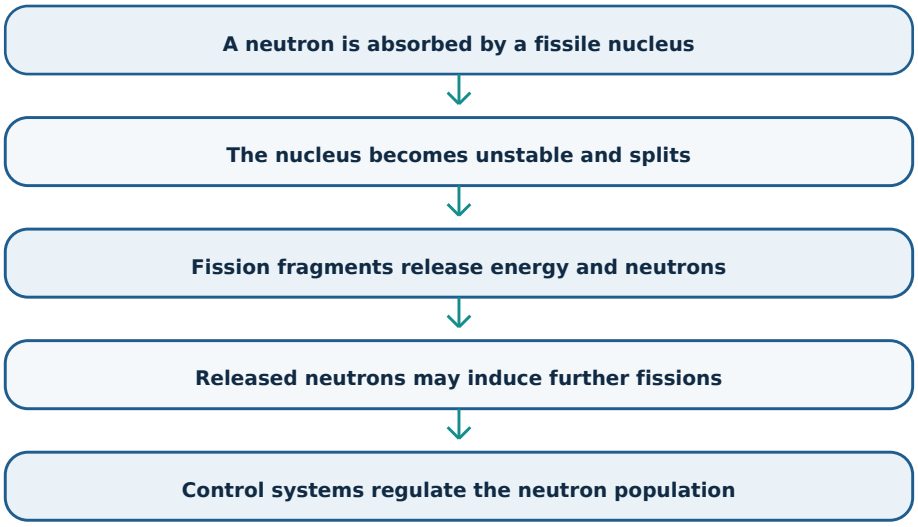
5.1 Nuclear energy and binding energy

Nuclear energy is associated with the structure and binding energy of atomic nuclei. A small change in nuclear mass can correspond to a large amount of energy, described by Einstein's relation $E = mc^2$. In electricity generation, the most established nuclear process is fission of heavy nuclei such as uranium-235 or plutonium-239.

5.2 Nuclear fission

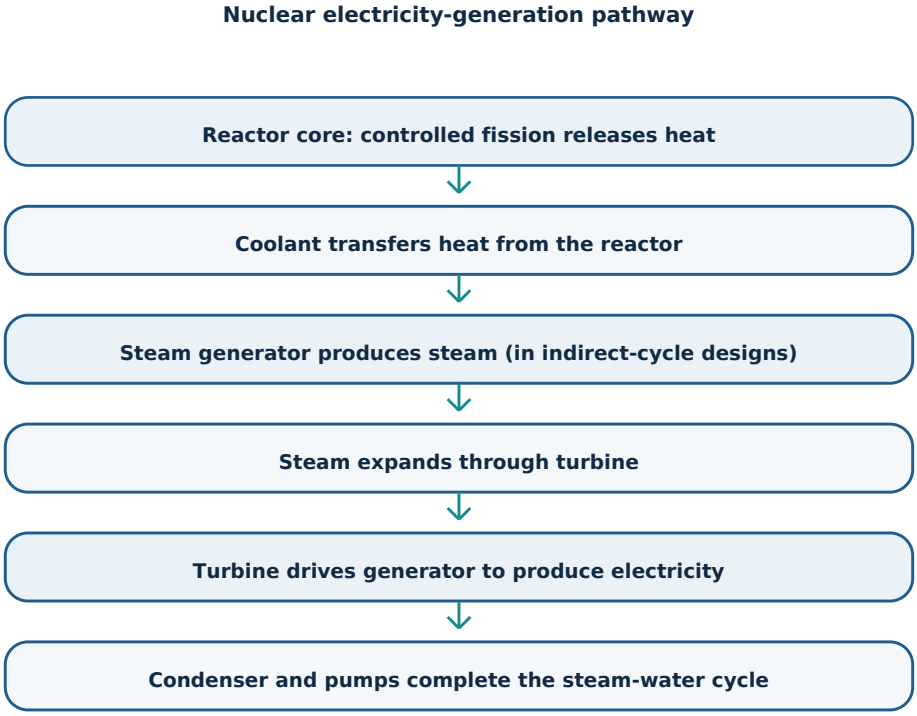
Fission occurs when a heavy nucleus absorbs a neutron and becomes unstable, then splits into two or more lighter nuclei. The reaction releases energy and additional neutrons. The energy appears mainly as kinetic energy of the fission fragments and is transferred as heat to the reactor coolant.

Simplified fission chain reaction



A chain reaction is self-sustaining when neutrons from one generation produce an appropriate number of fissions in the next generation. In a power reactor, the chain reaction is controlled. Control rods absorb neutrons; moderators in many reactor designs slow neutrons; coolant removes heat; shielding reduces radiation exposure; and engineered barriers contain radioactive materials.

5.3 Nuclear power station



The nuclear reactor is the heat source; the turbine-generator and steam cycle perform the conversion to electricity. Different reactor designs arrange coolant and steam circuits differently. The diagram is therefore a simplified teaching representation, not a detailed plant schematic.

5.4 Nuclear fusion

Fusion combines light nuclei to form a heavier nucleus, releasing energy when the products have greater binding energy per nucleon. A commonly studied reaction combines deuterium and tritium to produce helium-4, a neutron and energy. Fusion fuel must be heated to extremely high temperatures and confined long enough for fusion reactions to occur. Controlled fusion for commercial electricity remains a major engineering and scientific challenge.

$$^{2}H + {}^{3}H \rightarrow {}^{4}He + n + \text{energy}$$

5.5 Advantages and challenges

Potential advantages	Challenges and risks
High energy density and small fuel mass	High capital cost and long project development
Low direct carbon emissions during operation	Radioactive waste handling and long-term management
Can provide firm electricity independent of weather	Complex safety, cooling and radiation-protection systems
Small land footprint per unit of generation in many designs	Decommissioning, emergency planning and proliferation concerns

Nuclear engineering requires a safety culture based on defence in depth: multiple independent and diverse barriers prevent or mitigate failures. Plant siting, seismic design, cooling reliability, operator training, emergency preparedness, safeguards and waste management are essential considerations.

Unit 5 — check your understanding

- Distinguish fission from fusion.
- Explain why a reactor chain reaction must be controlled.

- Identify the function of control rods, moderator, coolant and shielding.
- Why does “low operational carbon emissions” not mean “zero environmental impact”?

UNIT 6 — SOLAR ENERGY

Unit purpose. Understand solar radiation, photovoltaic conversion, PV system components, solar thermal technologies and the calculations used to estimate power output.

6.1 Solar resource and technology families

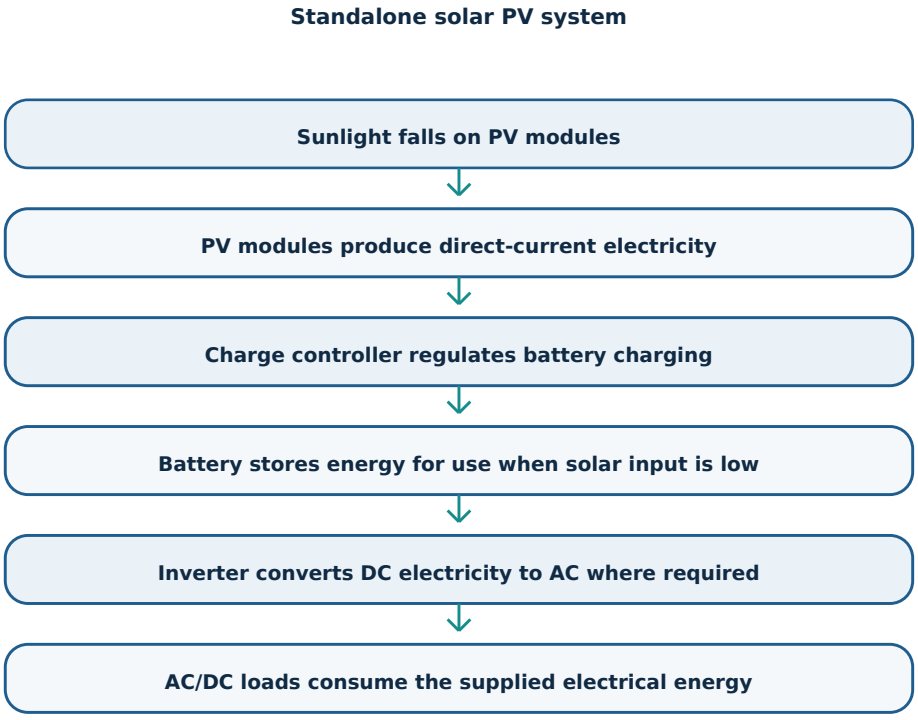
Solar energy is electromagnetic radiation from the Sun. At the Earth’s surface, the usable resource depends on latitude, season, time of day, cloud cover, atmospheric conditions, shading, surface orientation and local weather. Solar technologies are commonly grouped into photovoltaic systems, which convert light directly into electricity, and solar thermal systems, which capture radiation as heat.

6.2 Photovoltaic effect

A photovoltaic cell is a semiconductor device. When photons with sufficient energy are absorbed, they create charge carriers. The electric field associated with the cell junction helps separate these charges, producing a voltage. When an external circuit is connected, current flows and electrical power is delivered. Silicon is the dominant commercial PV material; thin-film and emerging materials are also researched and deployed.

Cells are interconnected to form modules; modules are connected into strings and arrays to achieve the required voltage and power. Module output depends on irradiance, cell temperature, shading, orientation, soiling, electrical mismatch and conversion equipment.

6.3 Main components of a standalone PV system



The PV array generates DC electricity. A charge controller manages battery charging and protects against unsuitable operating conditions. Batteries shift energy from periods of generation to periods of demand. An inverter converts DC to AC for conventional AC appliances. Protection devices, isolators, earthing, cables, mounting structures and monitoring equipment are also required. A grid-connected system may omit batteries where grid export/import and applicable rules make this appropriate.

6.4 Solar irradiance and energy

Irradiance is instantaneous solar power per unit area, commonly measured in W/m². Irradiation is solar energy received per unit area over a period, commonly measured in kWh/m²/day. The distinction matters: power is a rate; energy accumulates over time.

Incident solar power: $P_s = G\,A$ | Approximate electrical output: $P_e = \eta\,G\,A$

G is irradiance (W/m²), A is module area (m²) and η is conversion efficiency as a decimal. The simplified equation assumes uniform irradiance and a stated module efficiency; actual field output is reduced by temperature, wiring, mismatch, soiling, inverter and other system losses.

Worked Example 6.1 — PV module output

Problem. A PV module has an active area of 10 m², receives irradiance of 800 W/m² and has a conversion efficiency of 20%. Estimate its electrical power output under these conditions.

Solution.

Step 1. Incident solar power = G × A = 800 × 10 = 8,000 W.

Step 2. Convert efficiency to decimal: 20% = 0.20.

Step 3. Electrical output = 0.20 × 8,000 W = 1,600 W.

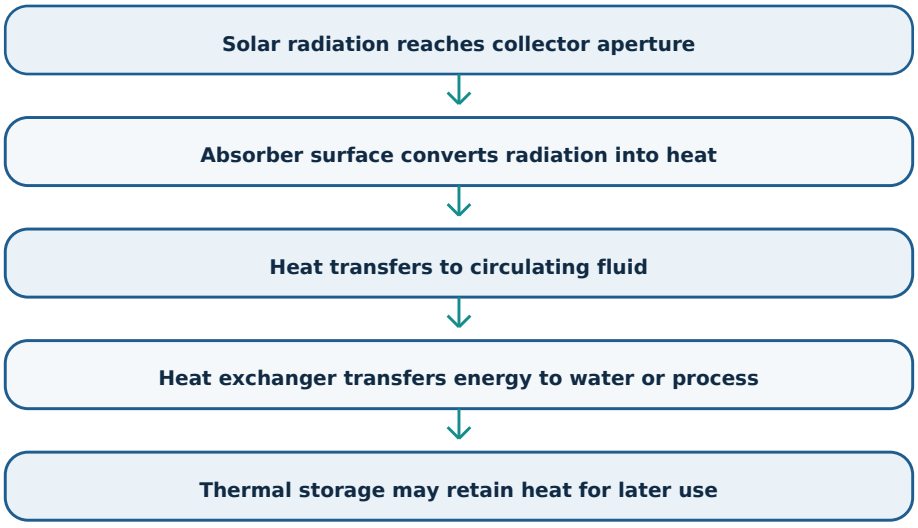
Step 4. Convert to kilowatts: 1,600 W ÷ 1,000 = 1.6 kW.

Final answer: 1.6 kW (idealised estimate).
Engineering meaning: This is a point-in-time estimate at the stated irradiance. Daily energy requires integrating output over the solar day and accounting for system losses.

6.5 Solar thermal systems

Solar thermal equipment captures radiation as heat. A flat-plate collector typically uses an absorber plate, glazing, insulation and fluid passages. An evacuated-tube collector reduces convective heat losses. The heated fluid can supply domestic hot water, process heat or space heating. Solar dryers and cookers use captured heat directly. Concentrating solar power systems use mirrors or lenses to focus sunlight and produce high-temperature heat, which can drive a thermodynamic power cycle.

Solar thermal heating pathway



6.6 Advantages, limitations and maintenance

- Advantages: modularity, no fuel purchase for sunlight, low operating emissions and suitability for rooftops, mini-grids and utility-scale plants.
- Limitations: output varies with time and weather; no direct generation at night; storage and inverter costs; land and material requirements.
- Maintenance: inspect modules and mounting, remove excessive soiling safely, check connectors and cables, monitor inverter alarms, test protection systems and follow battery manufacturer requirements.

Unit 6 — applied design note

Do not size a PV array from appliance wattage alone. A design requires daily energy demand (kWh/day), solar resource (peak-sun-hours or irradiation), system losses, autonomy requirement, battery chemistry and usable depth of discharge, inverter surge capacity, and the load profile. The simplified sizing example later in this document illustrates the energy balance.

UNIT 7 — BIOMASS ENERGY

Unit purpose. Examine organic feedstocks, conversion routes, biogas production, engineering constraints and sustainability conditions.

7.1 Meaning and sources of biomass

Biomass is organic material that can be used as an energy resource. Examples in the supplied lecture note include wood, agricultural residues, animal waste, municipal organic waste, energy crops, sawdust, rice husks and palm residues. Biomass stores chemical energy through photosynthesis. It can be used directly as a solid fuel or converted into gaseous and liquid fuels.

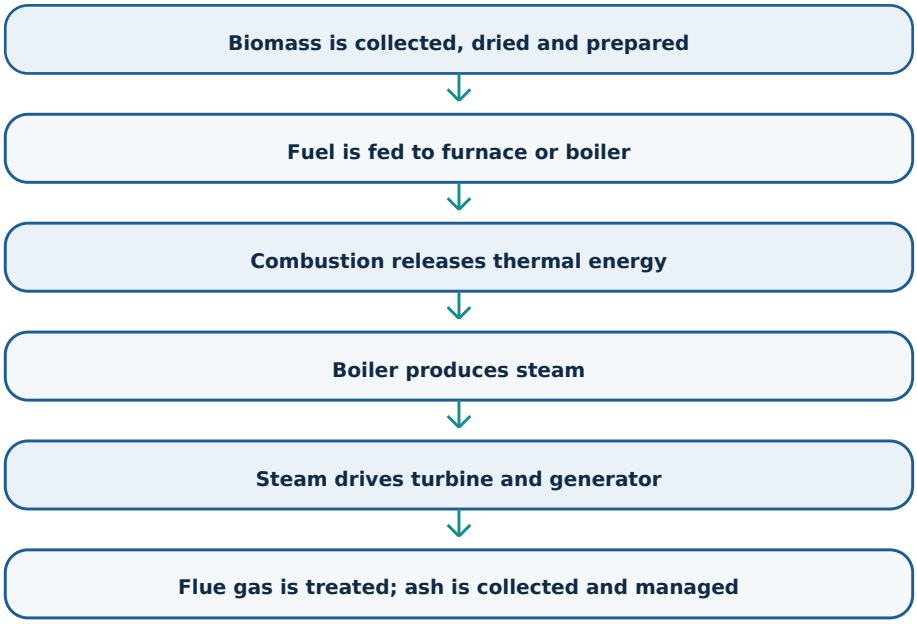
7.2 Biomass conversion pathways

Route	Basic process	Typical product / use
Combustion	Oxidise biomass with sufficient air	Heat; steam and electricity in a power cycle
Gasification	Partial oxidation at high temperature with limited oxygen/steam	Producer gas/syngas for heat, engines or further synthesis
Anaerobic digestion	Microorganisms break down organic matter without oxygen	Biogas rich in methane and carbon dioxide
Pyrolysis	Heat biomass in little or no oxygen	Bio-oil, char and gas, depending on process conditions
Fermentation	Microorganisms convert suitable sugars/starches	Alcohol fuels such as ethanol

7.3 Combustion and biomass power

In direct combustion, prepared biomass is burned in a furnace or boiler. Heat may be used directly or converted into steam to drive a turbine-generator. Feedstock moisture is important: wet biomass consumes heat to evaporate water, reducing useful heat available. Particle size, ash chemistry, fuel consistency, combustion-air distribution and emissions control influence performance.

Biomass combustion power pathway



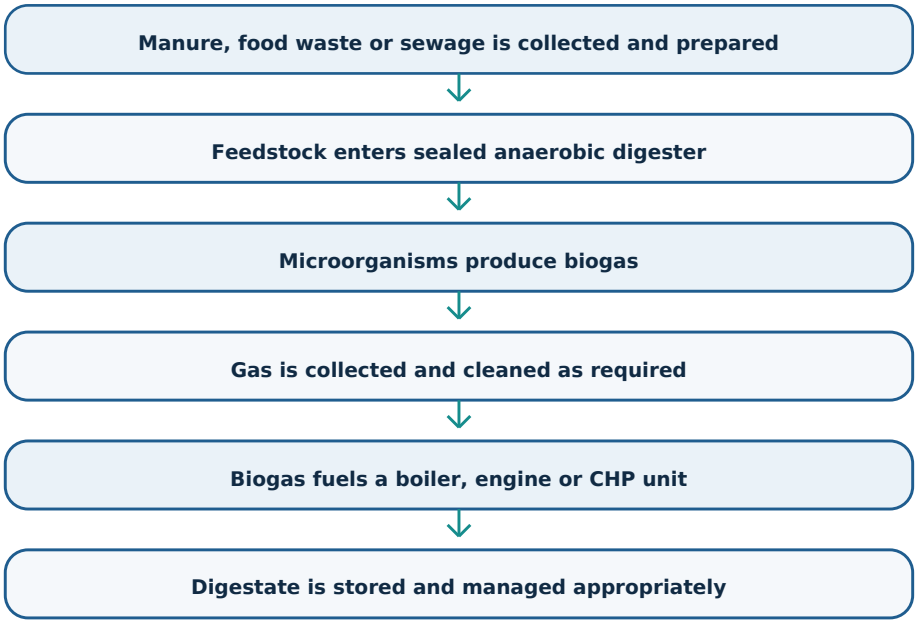
7.4 Gasification

Gasification converts carbonaceous feedstock into a combustible gas mixture under controlled partial oxidation. The gas may contain carbon monoxide, hydrogen, methane, carbon dioxide, nitrogen and contaminants such as tar and particulates. Before use in an engine or turbine, the gas may require cooling, filtration and contaminant removal. Gasifier type and feedstock characteristics determine gas quality.

7.5 Anaerobic digestion and biogas

Anaerobic digestion is the biological breakdown of organic material by microorganisms in the absence of oxygen. It proceeds through interacting stages commonly described as hydrolysis, acidogenesis, acetogenesis and methanogenesis. The resulting biogas contains methane and carbon dioxide, with smaller amounts of other gases depending on feedstock and process. Digestate remains after digestion and may be managed as a nutrient-containing material when quality and regulations permit.

Biogas production and use



7.6 Advantages and sustainability concerns

- Potential benefits: uses organic residues, can provide dispatchable energy, supports rural enterprise, may reduce uncontrolled waste disposal and can produce useful digestate.
- Challenges: seasonal feedstock supply, collection and transport cost, moisture, contamination, gas cleaning, plant operation and maintenance.
- Biomass is not automatically carbon-neutral. Sustainability depends on land-use change, feedstock regrowth, soil-carbon effects, transport, processing energy, methane leakage and what would otherwise have happened to the residue.

Worked Example 7.1 — electrical output from biomass

Problem. A plant receives 2,000 kg/h of dry biomass with a heating value of 15 MJ/kg. Overall electrical efficiency is 25%. Estimate electrical output.

Solution.

Step 1. Thermal energy input per hour = mass flow × heating value = 2,000 × 15 = 30,000 MJ/h.

Step 2. Convert energy rate to MW: 1 MW = 3,600 MJ/h, so input power = 30,000 ÷ 3,600 = 8.333 MW.

Step 3. Electrical output = efficiency × input power = 0.25 × 8.333 = 2.083 MW.

Final answer: Approximately 2.08 MW electrical.
Engineering meaning: The estimate assumes the stated dry-fuel heating value and overall efficiency apply continuously; auxiliary loads and real plant variability may change net export.

UNIT 8 — WIND ENERGY

Unit purpose. Apply the kinetic-energy principle to moving air, understand turbine components and use the wind-power equation to estimate available and extracted power.

8.1 Wind as a kinetic energy resource

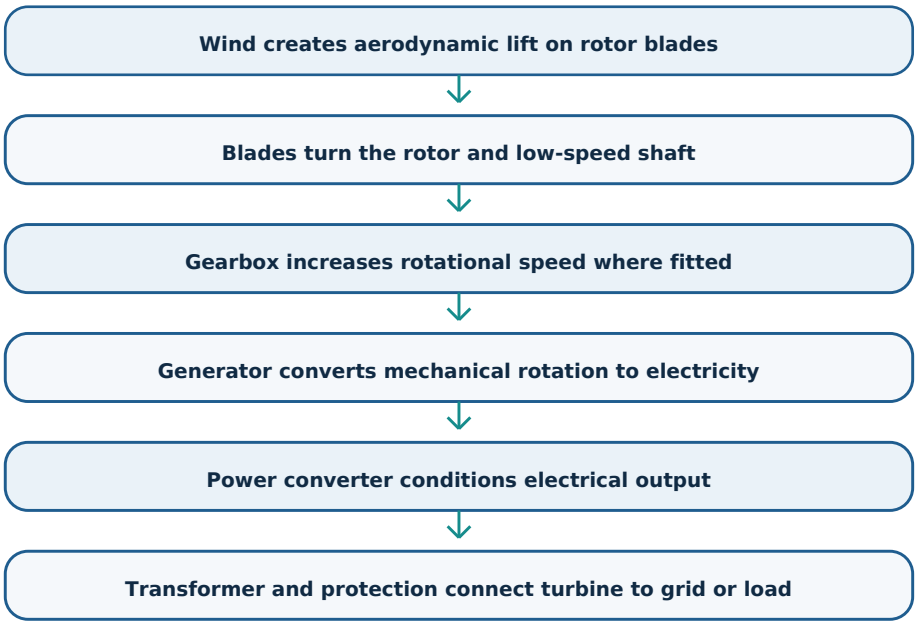
Wind is moving air and therefore contains kinetic energy. The power passing through a rotor swept area depends on air density, swept area and the cube of wind speed. This cubic relationship makes wind speed measurement and site selection especially important.

Available wind power: $P_{wind} = \frac{1}{2} \rho A v^3$

ρ is air density (kg/m³), A is rotor swept area (m²), and v is wind speed (m/s). For a circular rotor, $A = \pi D^2/4 = \pi R^2$. Because power varies with v^3, a 10% increase in wind speed produces a substantially larger increase in theoretical available power, all else equal.

8.2 Wind turbine components and operation

Horizontal-axis wind turbine power train



The rotor captures part of the kinetic energy in the wind. Pitch control changes blade angle to regulate power and protect the turbine. Yaw control turns the nacelle so the rotor faces the wind. A gearbox is used in some turbine designs to match rotor speed to generator requirements; direct-drive machines omit the gearbox. Modern turbines include brakes, sensors, converters, transformers and supervisory control systems.

8.3 Power coefficient and Betz limit

A turbine cannot extract all the kinetic energy from the air passing through the rotor. If it removed all the energy, the air would stop behind the rotor and no further air could pass through. The power coefficient Cp is the fraction of available wind power captured by the rotor. The theoretical Betz limit for an ideal actuator disk is approximately 0.593. Real turbines operate below this limit because of aerodynamic, mechanical and electrical losses.

Extracted mechanical power: $P_m = C_p \times \frac{1}{2} \rho A v^3$

8.4 Tip-speed ratio

$$\lambda = (\omega R) / v$$

Here ω is rotor angular speed in rad/s, R is rotor radius in metres and v is wind speed in m/s. Tip-speed ratio compares blade-tip speed with incoming wind speed. Turbine aerodynamic efficiency depends on operating near an appropriate tip-speed ratio; the optimum varies with rotor design.

Worked Example 8.1 — wind turbine power

Problem. A wind turbine has rotor diameter 20 m. Wind speed is 8 m/s, air density is 1.225 kg/m³ and Cp is 0.40. Estimate the mechanical power extracted.

Solution.

Step 1. Swept area $A = \pi D^2/4 = \pi(20^2)/4 = 314.16$ m².

Step 2. Available wind power = $0.5 \times 1.225 \times 314.16 \times 8^3$.

Step 3. Since $8^3 = 512$, available power ≈ 98,425 W.

Step 4. Extracted mechanical power = Cp × available power = 0.40 × 98,425 ≈ 39,370 W.

Step 5. Convert to kW: 39,370 ÷ 1,000 ≈ 39.37 kW.

Final answer: Approximately 39.4 kW mechanical.
Engineering meaning: This is an idealised operating-point estimate. Electrical output will be lower after drivetrain and generator/converter losses; turbine controls may also limit power.

8.5 Site assessment and operating limits

Wind projects require long-term wind measurements at or near hub height, assessment of turbulence and extreme winds, terrain and obstacle analysis, grid access, transport logistics, environmental screening and community engagement. A turbine has cut-in, rated and cut-out wind speeds. Below cut-in speed it does not generate; above rated speed controls limit output; at excessive wind speed it shuts down to protect the machine.

UNIT 9 — HYDROPOWER

Unit purpose. Explain the conversion of hydraulic energy into electricity, calculate plant output from flow and head, and distinguish major hydropower configurations.

9.1 Hydraulic energy and power

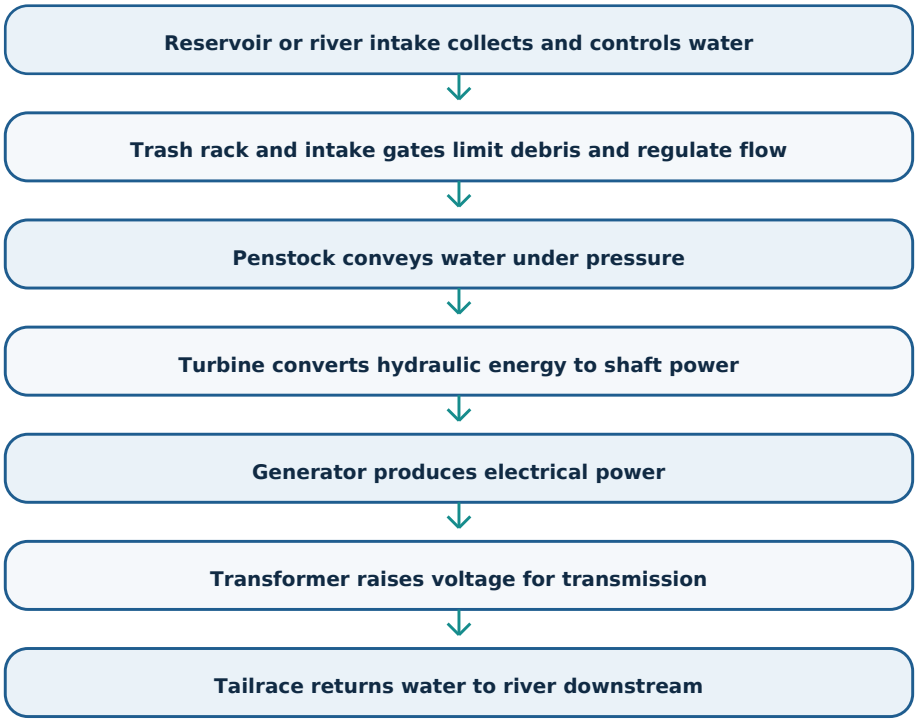
Hydropower uses the energy of water, principally the gravitational potential energy associated with elevation and, in some systems, the kinetic energy of flowing water. A hydroelectric plant uses hydraulic head to drive water through a turbine. The turbine converts hydraulic power to shaft power; the generator converts shaft power to electricity.

Ideal hydraulic power: $P_h = \rho g Q H$ | Electrical output: $P_{out} = \rho g Q H \eta$

ρ is water density (kg/m³), g is gravitational acceleration (m/s²), Q is flow rate (m³/s), H is net head (m) and η is the combined turbine-generator efficiency. Net head is the effective head available after hydraulic losses in intake, penstock, bends, valves and other conveyance components.

9.2 Main components of a hydroelectric plant

Hydroelectric plant — principal components



The intake controls the water entering the conveyance system. The penstock is designed for pressure, water hammer, structural loading and corrosion. Turbine selection depends on head and flow: impulse turbines such as Pelton are suited to high head and lower flow; reaction turbines such as Francis and Kaplan are used across other head/flow ranges. The generator and electrical system must match speed, grid frequency and operating requirements.

9.3 Types of hydropower

Type	Description	Typical engineering issue
Impoundment	A dam creates a reservoir and regulates water release	Reservoir footprint, inundation, sediment, ecology and resettlement
Run-of-river	Uses river flow with limited storage	Output follows river flow and seasonal hydrology
Pumped storage	Pumps water to an upper reservoir when electricity is available and generates during demand peaks	Round-trip losses, reservoir siting and high civil works cost
Small / micro hydro	Smaller installations serving local loads or mini-grids	Local flow data, maintenance capability and seasonal reliability

Worked Example 9.1 — hydroelectric output

Problem. A plant has flow rate 10 m³/s, net head 50 m, water density 1,000 kg/m³, gravitational acceleration 9.81 m/s² and overall efficiency 85%. Calculate electrical output.

Solution.

Step 1. Use Pout = ρ g Q H η.

Step 2. Substitute: Pout = 1,000 × 9.81 × 10 × 50 × 0.85.

Step 3. First calculate ideal hydraulic power: 1,000 × 9.81 × 10 × 50 = 4,905,000 W.

Step 4. Apply efficiency: 4,905,000 × 0.85 = 4,169,250 W.

Step 5. Convert to MW: 4,169,250 ÷ 1,000,000 = 4.169 MW.

Final answer: Approximately 4.17 MW.
Engineering meaning: The result assumes the stated net head and flow are available at the same operating point. Seasonal river flow may reduce output.

9.4 Small and micro hydropower

Small hydro can provide local electricity where river flow and head are suitable. Classification thresholds differ among countries and agencies, so “micro”, “mini” and “small” should not be treated as universal capacity categories. A feasibility study needs flow-duration curves, head survey, sediment and debris data, ecological flow requirements, access, civil works, turbine choice, grid or mini-grid design and maintenance planning.

9.5 Environmental and social considerations

Hydropower can alter river flow, sediment transport, fish migration, water temperature and downstream habitats. Reservoirs may inundate land and affect livelihoods. Environmental design can include environmental flows, fish passage where appropriate, sediment management, careful siting and meaningful community consultation.

UNIT 10 — GEOTHERMAL AND OTHER RENEWABLE ENERGY

Unit purpose. Introduce geothermal heat extraction and ocean-energy resources, and compare the operating characteristics of major renewable technologies.

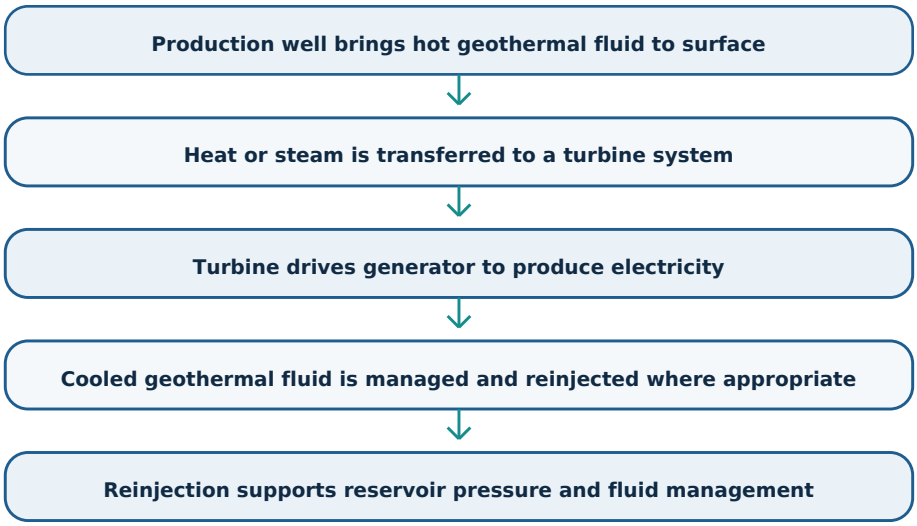
10.1 Geothermal energy

Geothermal energy is heat from within the Earth. It originates from residual heat from planetary formation and radioactive decay within the Earth. The accessible resource depends on local geology, temperature, depth, permeability and the presence of suitable fluids. Geothermal energy can be used directly for heating or indirectly to generate electricity.

10.2 Geothermal power-plant configurations

Plant type	Operating principle	Typical resource condition
Dry steam	Natural steam from a geothermal reservoir drives a turbine directly	Steam-dominated reservoir
Flash steam	High-pressure hot water is brought to lower pressure so part flashes to steam	High-temperature liquid-dominated resource
Binary cycle	Geothermal fluid heats a secondary working fluid with a lower boiling point; secondary vapour drives turbine	Moderate-temperature resource where direct steam is unsuitable

Simplified geothermal power pathway



A geothermal project must manage scaling, corrosion, dissolved gases, reservoir pressure and reinjection. Drilling is capital-intensive and resource confirmation is essential. Geothermal plants can provide relatively continuous output where the resource is suitable, but they are highly site-dependent.

10.3 Direct-use geothermal applications

Where temperatures are appropriate, geothermal heat can be used directly for district heating, greenhouses, aquaculture, bathing, drying and industrial processes. Direct-use applications may be more efficient than generating electricity from a low-temperature resource because they avoid a thermodynamic conversion stage.

10.4 Ocean energy

Ocean energy includes tidal range, tidal-stream, wave energy and ocean thermal energy conversion (OTEC). Tidal range uses the difference in water level between high and low tide. Tidal-stream systems extract kinetic energy from moving tidal currents. Wave systems capture energy from surface-wave motion. OTEC uses the temperature difference between warm surface water and colder deep water to operate a heat engine. These technologies face challenges involving marine corrosion, biofouling, storms, maintenance access, grid connection and cost.

10.5 Comparative characteristics

Technology	Primary resource	Strength	Main constraint
Solar PV	Solar radiation	Modular and widely deployable	Day/night and weather variability
Solar thermal	Solar radiation as heat	Direct useful heat; thermal storage possible	Weather and temperature limitations
Wind	Kinetic energy of moving air	Low fuel cost and scalable	Wind variability and site dependence
Hydro	Water head and flow	Can be controllable where storage exists	Hydrology, civil works and ecological impacts
Biomass	Chemical energy in organic feedstock	Dispatchable when fuel supply is secure	Feedstock logistics and sustainability
Geothermal	Earth heat	Can provide steady output at suitable sites	Exploration and drilling risk
Ocean	Tides, waves and ocean temperature gradients	Some resources are predictable	Technology maturity, marine environment and cost

Unit 10 — engineering selection

A technology should be selected by matching the resource to the required service. A coastal site with strong tidal currents may still be unsuitable if marine access and grid connection are poor. A geothermal resource may be excellent for direct heating but not hot enough for economical electricity generation. Resource quality, project scale, load profile, environmental effects and lifecycle cost must be assessed together.

UNIT 11 — ENERGY CONVERSION SYSTEMS

Unit purpose. Analyse conversion stages, calculate combined efficiency and understand why system boundaries and losses matter.

11.1 Energy conversion

Energy conversion is the process of changing energy from one form to another. Examples include chemical-to-thermal conversion in combustion, thermal-to-mechanical conversion in a turbine, mechanical-to-electrical conversion in a generator, and solar-radiant-to-electrical conversion in a photovoltaic cell. Most practical systems contain several conversion stages.

11.2 Efficiency of one conversion stage

$$\eta = \text{Useful output} / \text{Input}$$

Efficiency is dimensionless and is commonly stated as a percentage. The numerator and denominator must refer to compatible quantities and the same system boundary. For example, generator efficiency is electrical output divided by mechanical shaft input; whole-plant efficiency may be net electrical output divided by fuel energy input.

11.3 Overall efficiency of cascaded stages

When stages are connected in series, the overall efficiency is the product of the individual efficiencies, provided the output of one stage is the input to the next and the same energy path is being followed.

$$\eta_{overall} = \eta_1 \times \eta_2 \times \eta_3 \times \ldots \times \eta_n$$

Worked Example 11.1 — cascaded conversion efficiency

Problem. A renewable-energy system has turbine efficiency 90%, generator efficiency 95% and power-converter efficiency 96%. Determine overall efficiency.

Solution.

Step 1. Convert percentages to decimals: 0.90, 0.95 and 0.96.

Step 2. Multiply the stage efficiencies: $\eta_{overall} = 0.90 \times 0.95 \times 0.96$.

Step 3. $\eta_{overall} = 0.8208$.

Step 4. Convert to percentage: $0.8208 \times 100\% = 82.08\%$.

Final answer: 82.08%.
Engineering meaning: The overall system delivers about 82.08% of the energy entering the first stage as useful output at the final stage; the remainder is lost or consumed along the chain.

11.4 Comparing conversion pathways

System	Input form	Main conversion sequence	Output
PV	Solar radiation	Semiconductor photovoltaic effect	DC electricity
Wind	Air kinetic energy	Rotor → shaft → generator → converter	Electricity
Hydro	Water hydraulic energy	Turbine → shaft → generator	Electricity
Coal	Fuel chemical energy	Combustion → steam → turbine → generator	Electricity
Gas turbine	Fuel chemical energy	Combustion → hot gas expansion → generator	Electricity
Biomass	Feedstock chemical energy	Combustion or biochemical conversion	Heat, gas or electricity
Geothermal	Earth heat	Heat transfer/steam → turbine → generator	Electricity

11.5 System boundaries and parasitic loads

A plant may report gross generation at generator terminals and net generation after subtracting electricity used by pumps, fans, cooling systems, fuel handling and control equipment. Engineers must state whether efficiency is gross or net. A fair comparison also uses consistent fuel heating-value conventions, operating periods and system boundaries.

Unit 11 — practice

- A pump has 80% efficiency and a motor has 90% efficiency. Calculate combined efficiency.
- Explain why adding an inefficient conversion stage reduces overall efficiency.
- Distinguish generator efficiency from whole-power-station efficiency.

UNIT 12 — ENERGY STORAGE

Unit purpose. Explain why storage is required, classify major storage technologies and calculate nominal and usable battery energy.

12.1 Why energy storage is required

Electricity supply and demand must remain balanced in real time. Solar and wind output can vary with weather and time of day, while demand also changes. Storage shifts energy from one time to another, supports short-term balancing, provides backup, helps regulate frequency and voltage, reduces renewable curtailment and can improve resilience in off-grid systems.

12.2 Storage technology families

Family	Examples	How energy is stored
Electrochemical	Lead-acid, lithium-ion, sodium-ion, flow batteries	Reversible chemical reactions and electrochemical potential
Mechanical	Pumped hydro, compressed-air storage, flywheels	Gravitational potential, compressed-gas energy or rotational kinetic energy
Thermal	Hot water, molten salt, phase-change materials	Sensible or latent heat
Chemical	Hydrogen and synthetic fuels	Chemical energy in molecules produced using energy input

12.3 Battery capacity and usable energy

Battery capacity is often stated in ampere-hours (Ah), while nominal energy is estimated from nominal voltage and capacity. This simple calculation is an approximation because battery voltage changes during charge and discharge and the usable energy depends on chemistry, temperature, discharge rate, age and battery-management limits.

$$\text{Nominal battery energy (Wh)} \approx V \times Ah$$

Usable energy is lower than nominal energy when the design limits depth of discharge or includes conversion losses. If depth of discharge is expressed as a fraction, an initial estimate is $E_{usable} \approx E_{nominal} \times$ allowable depth of discharge. Inverter and wiring losses reduce the energy delivered to AC loads further.

Worked Example 12.1 — battery energy capacity

Problem. A battery bank is rated 24 V and 200 Ah. Calculate nominal energy. If 80% of nominal energy is considered usable, estimate usable energy before inverter losses.

Solution.

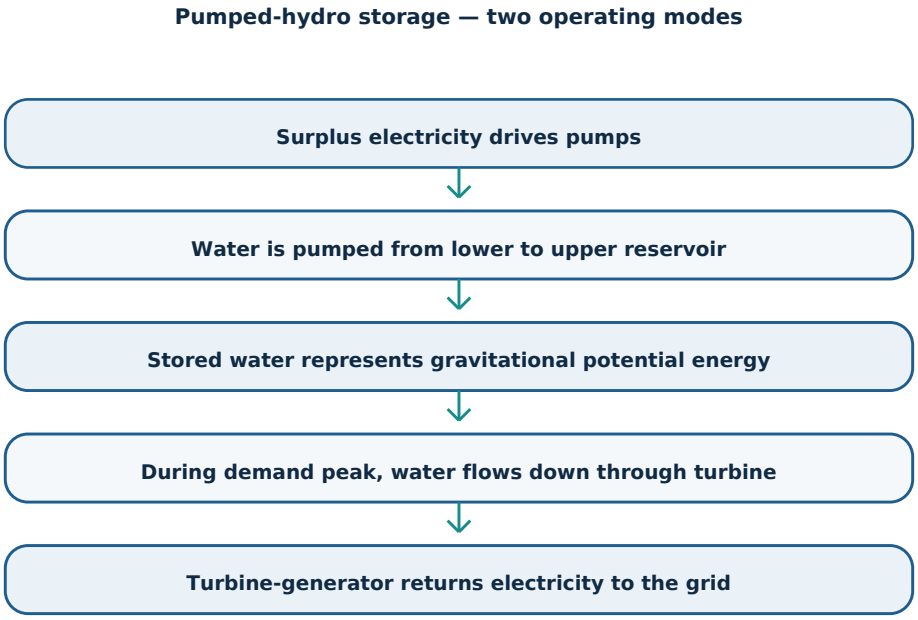
Step 1. Nominal energy = V × Ah = 24 × 200 = 4,800 Wh.

Step 2. Convert to kWh: 4,800 ÷ 1,000 = 4.8 kWh.

Step 3. Usable energy = 0.80 × 4.8 = 3.84 kWh.

Final answer: Nominal capacity ≈ 4.8 kWh; estimated usable energy ≈ 3.84 kWh.
Engineering meaning: The usable figure is a design estimate, not a guarantee. Battery datasheets and system operating limits must be used for final sizing.

12.4 Pumped-hydro storage



Pumped hydro is a large-scale mechanical storage method. During charging, pumps consume electricity to raise water. During discharge, water flows back through turbines to generate electricity. Round-trip efficiency is below 100% because of pump, turbine, generator and hydraulic losses. Site topography, reservoir availability, environmental impacts and civil works dominate project feasibility.

12.5 Hydrogen as an energy carrier

Renewable electricity can power an electrolyser to split water into hydrogen and oxygen. Hydrogen can be compressed or otherwise stored, transported and later used in a fuel cell, turbine or industrial process. Hydrogen is an energy carrier, not a primary energy source: energy must first be supplied to produce it. Conversion losses occur during electrolysis, compression, storage and reconversion.

12.6 Storage selection criteria

- Power rating (kW or MW): how quickly energy can be charged or discharged.
- Energy capacity (kWh or MWh): how much energy can be stored.
- Duration: how long the system can sustain its rated output.
- Round-trip efficiency: energy delivered divided by energy used to charge.
- Cycle life, calendar life, safety, temperature range and maintenance.
- Cost, siting, environmental footprint, recycling and end-of-life management.

UNIT 13 — ENERGY TRANSMISSION, CONSERVATION AND EFFICIENCY

Unit purpose. Understand why electricity is transmitted at high voltage, calculate three-phase current and resistive losses, and distinguish conservation from efficiency.

13.1 Electrical transmission

Electricity is generated at power stations and transferred through transmission networks to substations and distribution networks. Transmission commonly uses high voltage because, for a given power transfer, higher voltage allows lower current. Lower current reduces resistive heating losses in conductors.

For a balanced three-phase system, real power is related to line voltage, line current and power factor. The commonly used relationship is:

$$P = \sqrt{3}\, V_L\, I_L \cos \varphi$$

V_L is line-to-line RMS voltage, I_L is line current and cos φ is power factor. The factor √3 is essential in the standard balanced three-phase line-voltage form. Resistive loss in a conductor is I^2R; for a complete three-phase line, total loss depends on the resistance of all phase conductors and the current in each.

Worked Example 13.1 — three-phase line current

Problem. A balanced three-phase load receives 10 MW at 100 kV line-to-line with power factor 0.9. Calculate line current.

Solution.

Step 1. Use $P = \sqrt{3}\, V_L\, I_L \cos \varphi$.

Step 2. Rearrange: $I_L = P / (\sqrt{3}\, V_L \cos \varphi)$.

Step 3. Substitute: $I_L = 10{,}000{,}000 / (1.732 \times 100{,}000 \times 0.9)$.

Step 4. Denominator ≈ 155,880; current ≈ 64.15 A.

Final answer: Approximately 64.2 A.
Engineering meaning: The calculation assumes a balanced three-phase load and uses real power. For a detailed line-loss calculation, use the resistance per phase and account for the complete circuit.

13.2 Resistive transmission losses

$$\text{Conductor loss: } P_{loss} = I^2 R$$

Because losses vary with the square of current, doubling current increases resistive loss by a factor of four for the same resistance. This is one reason high-voltage transmission is used for bulk power transfer. In practice, line losses also depend on conductor temperature, length, geometry, reactive power, transformer losses and network operating conditions.

13.3 Energy conservation

Energy conservation means avoiding unnecessary energy use while maintaining the required service. Examples include switching off equipment when not required, reducing unnecessary lighting, improving building operation schedules and eliminating compressed-air leaks. Conservation often changes behaviour or operating practice.

13.4 Energy efficiency

Energy efficiency means delivering the same or better service with less energy input. Examples include replacing inefficient lamps with LEDs, using high-efficiency motors, improving insulation, selecting efficient pumps and fans, and using efficient transformers. Efficiency is generally assessed against a defined service or output.

Action	Category	Reason
Switch off lights in an unoccupied room	Conservation	Avoids unnecessary use
Replace a 100 W incandescent lamp with a 15 W LED of suitable light output	Efficiency	Provides comparable illumination with lower power
Repair compressed-air leaks	Both	Reduces wasted demand and improves system performance
Improve motor and pump matching	Efficiency	Reduces input energy for required flow or torque

13.5 Energy audits and management

An energy audit establishes where energy is used, identifies avoidable losses and evaluates improvement opportunities. A practical audit records utility bills, operating hours, equipment ratings, load profiles, process requirements and maintenance condition. Measures should be prioritised by energy savings, cost, reliability, safety, implementation time and effects on production or comfort.

UNIT 14 — ENERGY ECONOMICS AND ENERGY MIX

Unit purpose. Evaluate energy technologies beyond their physical performance by considering cost, risk, capacity factor, affordability and the value of a diversified energy supply.

14.1 Meaning of energy mix

An energy mix is the combination of energy sources used by a country, region, industry or facility. It may include fossil fuels, nuclear generation and renewable resources. A diversified mix can reduce dependence on one fuel or technology, improve resilience and balance different operating characteristics.

14.2 Technical factors in energy-mix analysis

- Installed capacity and expected annual generation.
- Resource availability, variability and seasonal patterns.
- Dispatchability, ramping capability and start-up constraints.
- Capacity factor and plant availability.
- Transmission access, grid stability and storage needs.
- Geographical distribution and exposure to climate or fuel-supply risk.

14.3 Economic factors

Energy project economics may include capital expenditure, operating and maintenance costs, fuel costs, financing costs, taxes, insurance, replacement components, decommissioning and environmental compliance. The correct comparison depends on project life, discount rate, capacity factor, fuel-price assumptions, degradation and system boundary.

14.4 Levelized cost of electricity (LCOE)

LCOE is a summary metric that expresses the discounted lifetime cost of a generating project per unit of electricity generated. It is useful for comparing technologies under stated assumptions, but it does not by itself capture all system value. It may not fully represent dispatchability, location, transmission upgrades, balancing, resilience, environmental externalities or the timing of generation.

LCOE = Present value of lifetime costs / Present value of lifetime electricity generated

Worked Example 14.1 — simplified LCOE

Problem. A solar project costs ₦100 million and is expected to generate 2,000 MWh each year for 20 years. Ignore discounting, replacement, operating and maintenance costs. Calculate the simplified cost per kWh.

Solution.

Step 1. Lifetime generation = 2,000 MWh/year × 20 years = 40,000 MWh.

Step 2. Convert to kWh: 40,000 MWh × 1,000 = 40,000,000 kWh.

Step 3. Simplified cost per kWh = ₦100,000,000 ÷ 40,000,000 kWh.

Step 4. Result = ₦2.50/kWh.

Final answer: ₦2.50/kWh under the deliberately simplified assumptions.
Engineering meaning: This is not a bankable LCOE. A real calculation includes discounting, annual O&M;, replacements, degradation, financing, taxes and actual energy yield.

14.5 Capacity factor

Capacity factor compares actual energy generated over a period with the energy that would have been generated if the plant operated at rated capacity for every hour in that period. It is a measure of utilisation over time, not a direct measure of conversion efficiency.

Capacity factor = Actual energy generated / (Rated capacity × Hours in period)

Worked Example 14.2 — annual capacity factor

Problem. A 5 MW plant generates 8,000 MWh in one year. Calculate its capacity factor using 8,760 hours per year.

Solution.

Step 1. Maximum possible generation = 5 MW × 8,760 h = 43,800 MWh.

Step 2. Capacity factor = 8,000 ÷ 43,800 = 0.18265.

Step 3. Convert to percentage: 0.18265 × 100% = 18.26%.

Final answer: Approximately 18.3%.
Engineering meaning: A solar plant can have a lower capacity factor than a thermal plant because solar resource is unavailable at night; this does not by itself indicate poor module efficiency.

14.6 Payback, NPV and IRR

Simple payback is the time required for cumulative savings or cash inflows to equal the initial investment. It is easy to communicate but ignores the time value of money and cash flows after payback. Net present value (NPV) discounts future cash flows to present value and subtracts initial investment. Internal rate of return (IRR) is the discount rate at which NPV equals zero. These metrics require realistic assumptions about project life, financing, operating costs and risk.

14.7 Why a diversified energy mix matters

A robust energy mix balances affordability, reliability, emissions, resource availability and national development priorities. Solar and wind may reduce fuel consumption but require flexibility and grid integration. Hydropower can provide flexibility where water availability and environmental conditions permit. Biomass and gas may provide dispatchable output but depend on fuel supply and emissions management. Nuclear can provide firm low-operational-carbon generation but requires high institutional and capital capacity.

UNIT 15 — SUSTAINABLE ENERGY AND SUSTAINABLE DEVELOPMENT

Unit purpose. Evaluate energy systems against environmental, economic, social and technical criteria, and relate renewable energy to sustainable development in Nigeria.

15.1 Meaning of sustainability

Sustainability means meeting present needs without undermining the ability of future generations to meet their own needs. Sustainable energy therefore must be environmentally responsible, economically viable, socially inclusive and technically reliable. A technology should not be judged solely by whether its resource is renewable; its full lifecycle and effects on communities also matter.

15.2 Three pillars of sustainability

Pillar	Core concerns	Energy-sector examples
Environmental	Emissions, pollution, water, land, biodiversity and waste	Life-cycle emissions, river impacts, mining and recycling
Economic	Affordability, investment, jobs, operation and maintenance	Cost of electricity, local supply chains, financing and fuel risk
Social	Access, equity, health, safety, livelihoods and participation	Rural electrification, energy poverty, resettlement and community benefits

The three pillars are interdependent. A project may be technically efficient but socially unacceptable if it displaces communities without fair process. A low-cost project may be unsustainable if it causes severe pollution. A low-emission technology may fail if spare parts and maintenance skills are unavailable.

15.3 Environmental dimension

Sustainable energy planning seeks to reduce greenhouse-gas emissions, air and water pollution, land degradation, biodiversity loss and hazardous waste. Renewable technologies can reduce dependence on fossil fuels, but they also have environmental footprints: hydropower can alter rivers; solar farms require land and materials; wind projects can affect landscapes and wildlife; biomass can cause emissions and deforestation if poorly managed.

15.4 Economic dimension

A sustainable energy system must remain financially viable over its operating life. This requires realistic capital and operating budgets, affordable tariffs, access to finance, local technical capacity, spare parts, maintenance arrangements, fuel-price risk management and productive uses of energy. Employment and local manufacturing can improve development benefits when supply chains are designed responsibly.

15.5 Social dimension

Energy sustainability includes equitable access to modern energy services. Energy poverty affects health, education, productivity, safety and household time use. Rural electrification, clean cooking, reliable electricity for clinics and schools, affordability protections, worker safety and meaningful community participation are important social considerations.

15.6 Renewable energy and the Sustainable Development Goals

The supplied lecture note links renewable energy particularly to SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production) and SDG 13 (Climate Action). Reliable energy can also support education, healthcare, employment, poverty reduction and industrial development.

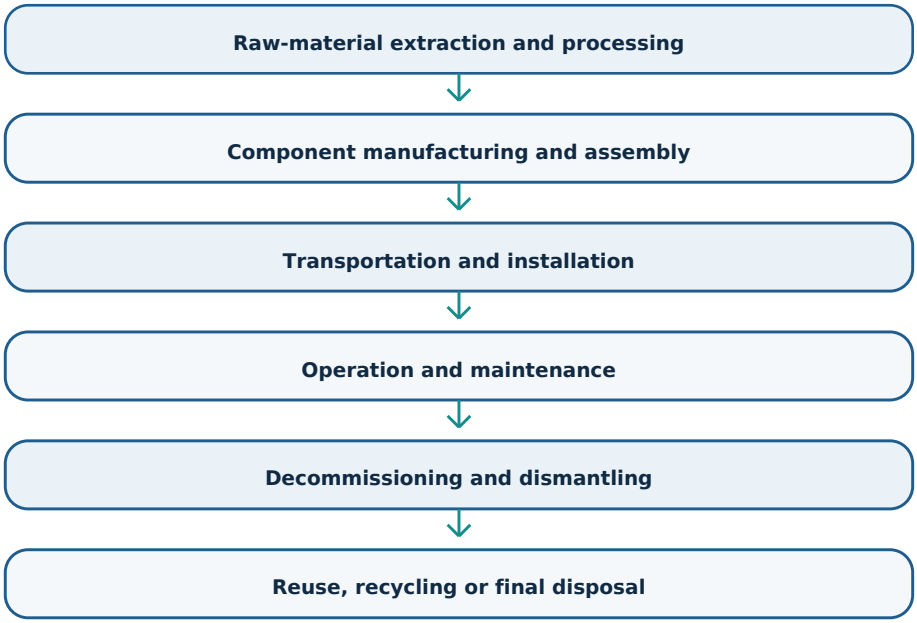
15.7 Sustainable energy in Nigeria

The supplied source frames Nigeria’s energy challenge around three linked priorities: energy access, economic development and climate/environmental responsibility. A credible transition pathway must improve reliability and affordability while expanding modern energy access and reducing environmental harm. This requires grid investment, distributed generation where appropriate, efficient appliances and industry, cleaner cooking, storage, skills development and institutions capable of planning and maintaining infrastructure.

15.8 Lifecycle assessment

Life-cycle thinking examines impacts from raw-material extraction through manufacturing, transport, installation, operation, maintenance and end-of-life treatment. It prevents a narrow comparison that counts only emissions at the power-plant gate. For example, PV modules have no combustion emissions during operation, but their production requires materials and energy; end-of-life collection and recycling should be considered.

Life-cycle stages of an energy technology



Unit 15 — evaluation exercise

A local government proposes a biomass plant using agricultural residues. Evaluate the project under all three sustainability pillars. Consider feedstock sustainability and competing uses, emissions control, transport distance, local employment, electricity affordability, digestate/ash management, community acceptance and long-term maintenance.

SUPPLEMENTARY TOPICS RETAINED FROM THE SUPPLIED LECTURE NOTE

The supplied 67-page source continues beyond the 15 three-hour teaching units. The following topics are retained because they extend the core course into integrated systems, planning, Nigerian energy issues and emerging grid technologies.

16. Comparative analysis of energy technologies

Technology comparisons should use consistent assumptions. Relevant dimensions include renewability, fuel requirement, operational emissions, dispatchability, storage need, fuel-price exposure, site dependence, waste and lifecycle impacts.

Technology	Renewable?	Dispatchability / variability	Key challenge
Coal	No	Generally dispatchable	High emissions and mining impacts
Natural gas	No	Generally dispatchable if fuel is available	Fuel price, methane leakage and emissions
Nuclear	No	Firm generation	Capital cost, safety and radioactive waste
Solar PV	Yes	Variable with sunlight	Storage, grid integration and land/materials
Wind	Yes	Variable with wind	Resource assessment and balancing
Hydro	Yes	Can be dispatchable if reservoir storage exists	Hydrology and ecosystem impacts
Biomass	Conditionally	Dispatchable with reliable feedstock	Sustainable feedstock and emissions
Geothermal	Yes	Can provide steady output	Site-specific resource and drilling risk

17. Energy end-use sectors and electrification

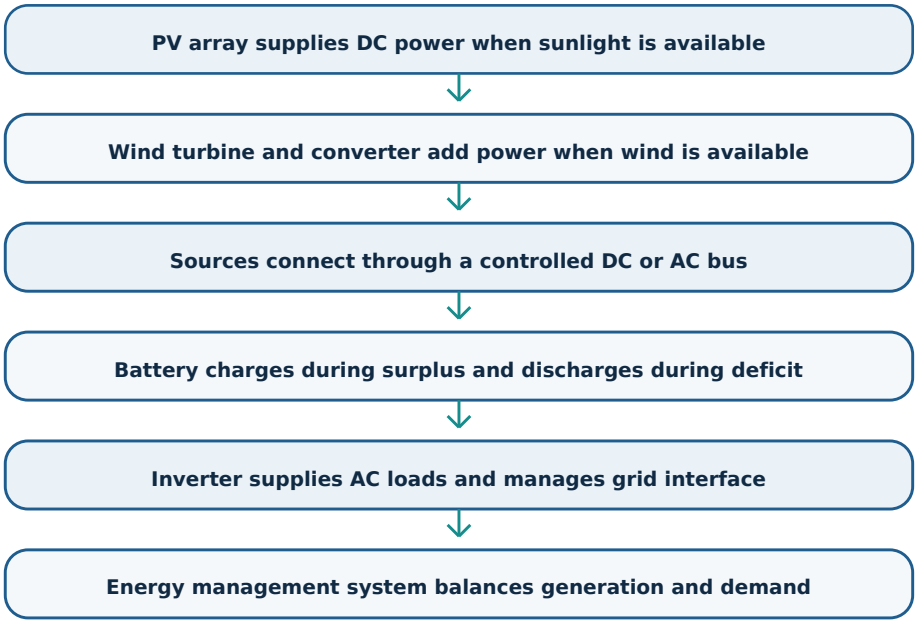
Residential end uses include lighting, refrigeration, cooking, cooling, water heating and electronics. Industrial end uses include motors, process heat, pumping, compressed air and production systems. Transport includes petrol/diesel vehicles, electric vehicles, hydrogen vehicles and rail. Agriculture uses energy for irrigation, drying, processing and refrigeration.

Electrification replaces direct use of fuels with electricity where technically and economically appropriate—for example, electric motors instead of combustion engines in some applications, heat pumps instead of some direct heating systems, and electric vehicles instead of internal-combustion vehicles. The climate and economic benefits depend on electricity generation mix, equipment efficiency, infrastructure and operating pattern.

18. Integrated and hybrid renewable-energy systems

A hybrid energy system combines two or more energy resources or technologies, often with storage and a common control system. Examples include solar plus battery, solar plus diesel, solar plus wind, solar plus hydro, solar plus biomass, and solar–wind–battery systems. Hybridisation can reduce dependence on one variable resource and improve supply reliability, but it increases control, protection and maintenance complexity.

Solar–wind–battery hybrid microgrid



Worked Example 18.1 — daily energy demand for a health centre

Problem. A rural health centre uses lighting at 1 kW for 6 h/day, a refrigerator at 0.5 kW for 10 h/day, a water pump at 2 kW for 2 h/day and communications equipment at 0.2 kW for 8 h/day. Calculate daily energy demand.

Solution.

Step 1. Lighting: 1 × 6 = 6 kWh/day.

Step 2. Refrigerator: 0.5 × 10 = 5 kWh/day.

Step 3. Pump: 2 × 2 = 4 kWh/day.

Step 4. Communications: 0.2 × 8 = 1.6 kWh/day.

Step 5. Total = 6 + 5 + 4 + 1.6 = 16.6 kWh/day.

Step 6. If system losses are 15%, required generation = 16.6 ÷ 0.85 = 19.53 kWh/day.

Final answer: Load demand = 16.6 kWh/day; estimated generation requirement = 19.53 kWh/day.
Engineering meaning: The 15% loss allowance is an assumed planning factor. Final sizing requires measured load profile, starting currents, critical-load priorities and resource data.

19. Energy planning and energy security

Energy planning determines how much energy is required, where it will come from, how it will be converted, transported and stored, what it will cost and what environmental and social effects may occur. Engineers should establish demand, local resource availability, technology suitability, capital and operating cost, reliability, infrastructure requirements and local maintenance capability.

Energy security means reliable access to sufficient, affordable and sustainable energy. It can be improved through diversified resources, domestic renewable energy, efficient infrastructure, storage, strong transmission networks, strategic fuel reserves and demand-side management.

20. Nigerian energy policy and implementation context

The supplied lecture note identifies inadequate electricity supply, transmission constraints, distribution losses, fossil-fuel dependence, rural access, cooking-energy challenges, unreliable supply, energy poverty, inadequate investment and infrastructure limitations as important issues. It identifies solar mini-grids and home systems, water pumping, telecommunications, small hydro, agricultural-residue biomass and selected wind applications as opportunities. Each opportunity requires site-specific feasibility work.

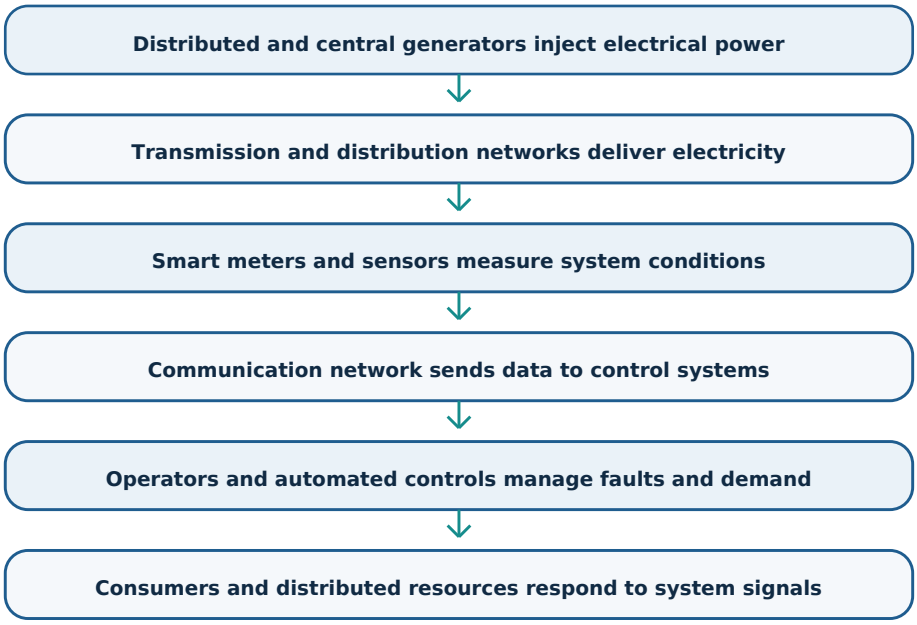
21. Energy transition: traditional and emerging systems

A traditional centralised pathway may be represented as fossil fuel → central power station → transmission → distribution → consumer. Emerging systems integrate central generation with distributed solar, wind, hydro, biomass, flexible generation, storage, smart controls, electric vehicles and demand response. The future system is therefore more interconnected and requires coordination between generation, networks, storage and consumers.

22. Smart energy systems and smart grids

A smart grid uses communication, measurement, automation and control to improve electricity-system operation. Sensors and smart meters provide information; communication networks transmit measurements; control centres and field devices support switching, fault detection, voltage management, demand response and renewable integration. Smart-grid benefits include improved monitoring, reliability, demand management, reduced losses and greater consumer participation. Cybersecurity, interoperability, privacy and communications resilience are essential design issues.

Simplified smart-grid information and power relationship



23. Environmental impacts and life-cycle thinking

Fossil-fuel impacts include carbon dioxide, methane, nitrogen oxides, sulphur oxides, particulate matter, oil spills, mining degradation and water contamination. Renewable systems also have impacts: solar materials and land use; wind visual, noise and wildlife interactions; hydropower habitat, sediment and displacement effects; biomass air emissions and land-use risks. A fair assessment uses lifecycle boundaries and transparent assumptions.

ENGINEERING FORMULA REFERENCE

Quantity	Formula	Notes / units
Energy from constant power	$E = P\,t$	P in W, t in s gives E in J; kW × h gives kWh
Kinetic energy	$E_k = \frac{1}{2} m v^2$	m in kg; v in m/s
Gravitational potential energy	$E_p = m g h$	m in kg; g in m/s²; h in m
Efficiency	η = useful output / input	Multiply by 100 for percentage
Solar electrical power estimate	$P = \eta G A$	G in W/m²; A in m²
Wind available power	$P = \frac{1}{2} \rho A v^3$	$A = \pi D^2/4$; v in m/s
Wind extracted power	$P_m = C_p \times \frac{1}{2} \rho A v^3$	Cp below Betz limit ≈ 0.593
Tip-speed ratio	$\lambda = \omega R / v$	ω in rad/s; R in m; v in m/s
Hydro output	$P = \rho g Q H \eta$	Q in m³/s; H in m
Battery nominal energy	$E \approx V \times Ah$	Wh; usable energy lower
Three-phase real power	$P = \sqrt{3}\, V_L I_L \cos\varphi$	Balanced system; line voltage/current
Resistive loss	$P_{loss} = I^2 R$	R in Ω; current in A
Capacity factor	CF = Eactual / (Prated × hours)	Use consistent units and period
Cascaded efficiency	$\eta_{overall} = \eta_1 \eta_2 \ldots \eta_n$	For series conversion stages
Simple energy requirement	$E = \Sigma(P_i\, t_i)$	Sum appliance energy by load

CALCULATION PRACTICE

Attempt each problem independently. Show the formula, substitutions, units and final interpretation. Unless stated otherwise, use g = 9.81 m/s², water density = 1,000 kg/m³ and air density = 1.225 kg/m³.

- A 5 kW motor operates for 8 hours per day. Determine daily energy consumption in kWh and MJ.
- A hydroelectric plant has flow rate 15 m³/s, head 40 m and efficiency 80%. Calculate electrical output.
- A wind turbine has rotor radius 15 m, wind speed 10 m/s and Cp = 0.38. Calculate extracted mechanical power.
- A PV array has area 20 m², irradiance 700 W/m² and efficiency 18%. Estimate output power.
- A 48 V, 300 Ah battery bank is used in an off-grid system. Calculate nominal energy capacity.
- A 10 MW generator produces 25,000 MWh annually. Calculate its capacity factor.
- A balanced three-phase line transfers 50 MW at 132 kV and power factor 0.95. Calculate line current.
- A biomass plant consumes 1,500 kg/h with heating value 18 MJ/kg and electrical efficiency 30%. Estimate output power.

SHORT-ANSWER REVISION QUESTIONS

- Define energy and distinguish it from power.
- Explain primary, secondary, final and useful energy with one example chain.
- List the main stages of a global energy system.

- Compare renewable and non-renewable energy resources.
- Describe oil, coal and natural-gas extraction pathways.
- Explain the operation of a coal-fired steam power station.
- Compare open-cycle and combined-cycle gas turbines.
- Explain nuclear fission and a controlled chain reaction.
- Describe the photovoltaic effect and the components of a standalone PV system.
- Distinguish solar thermal from solar photovoltaic technologies.
- Compare biomass combustion, gasification and anaerobic digestion.
- Derive the wind-power relationship and explain the Betz limit.
- Explain how flow rate and head affect hydropower output.
- Compare dry-steam, flash-steam and binary geothermal plants.
- Explain why cascaded conversion stages reduce overall efficiency.
- Compare electrochemical, mechanical, thermal and chemical storage.
- Explain why high-voltage transmission reduces resistive losses.
- Distinguish energy conservation from energy efficiency.
- Define LCOE and state two limitations of using it alone.
- Define capacity factor and explain why it is not the same as efficiency.
- Discuss the environmental, economic and social pillars of sustainability.
- Explain how hybrid systems and smart grids can support renewable integration.

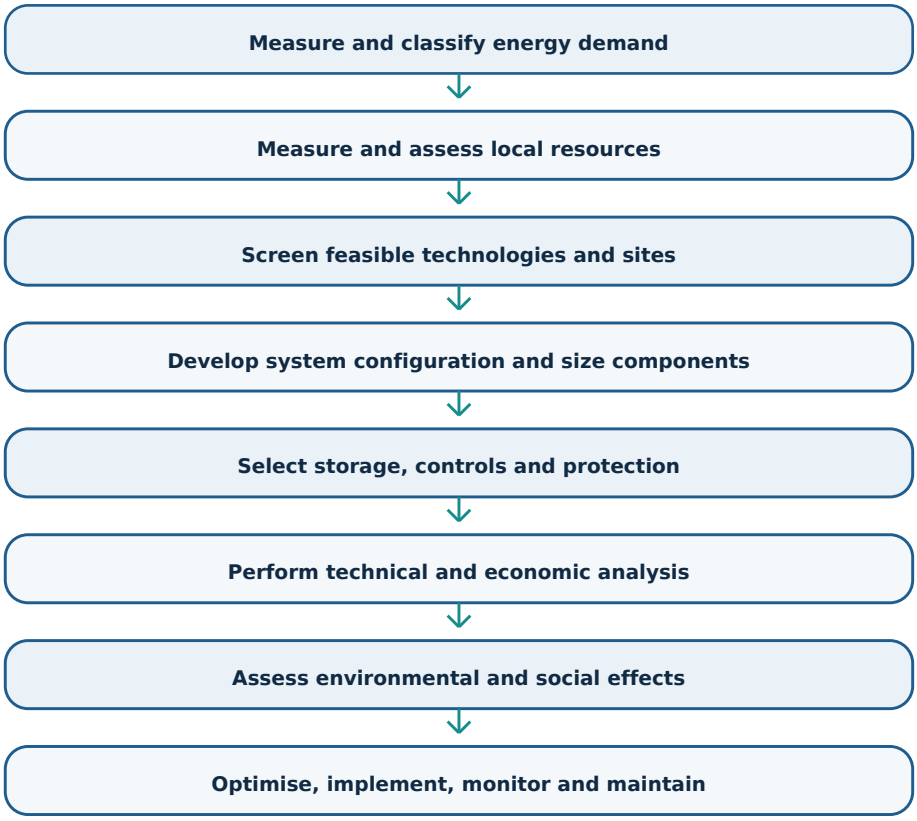
DISCUSSION AND DESIGN ASSIGNMENT

Scenario. A rural community requires reliable electricity for households, a health centre, schools, water pumping and small businesses. The available resources are strong solar radiation, moderate wind, a nearby river and agricultural residues.

Tasks. Assess daily and peak demand; identify critical loads; evaluate the local solar, wind, hydro and biomass resources; recommend a standalone, mini-grid or grid-connected arrangement; select storage; estimate generation and conversion losses; evaluate capital and operating costs; identify environmental and social impacts; and propose a maintenance and governance plan.

Recommended engineering workflow.

Technology-selection workflow



SELECTED REFERENCES AND TECHNICAL READING

The following sources are suitable starting points for checking technical principles, technology characteristics and current project data. The original supplied lecture note also lists standard textbooks; students should consult the editions available through their institution’s library.

- National Renewable Energy Laboratory (NREL), Renewable Energy: An Overview. A concise technology overview covering hydropower, bioenergy, geothermal, solar, wind and ocean energy.
- NREL, 2025 Annual Technology Baseline. Technology-specific cost and performance assumptions for generation and storage; use the assumptions and year stated in any comparison.
- U.S. Environmental Protection Agency, Renewable Energy Technology Basics. Overview of solar, wind and biomass technology fundamentals.
- International Renewable Energy Agency (IRENA), Renewable Capacity Statistics. Use the latest available edition when quoting installed renewable capacity; distinguish capacity from generation.
- Federal Government of Nigeria / Nigeria Energy Transition Office, Nigeria Energy Transition Plan. Consult the official document and subsequent policy updates for current targets and assumptions.
- Boyle, G. (Ed.), Renewable Energy: Power for a Sustainable Future. Oxford University Press.
- Twidell, J. and Weir, T., Renewable Energy Resources. Routledge.
- Duffie, J. A. and Beckman, W. A., Solar Engineering of Thermal Processes. Wiley.
- Masters, G. M., Renewable and Efficient Electric Power Systems. Wiley.
- Manwell, J. F., McGowan, J. G. and Rogers, A. L., Wind Energy Explained. Wiley.
- Kaltschmitt, M., Streicher, W. and Wiese, A., Renewable Energy: Technology, Economics and Environment. Springer.

Source and diagram note. The core course organisation, unit sequence, key subject areas and supplied numerical examples were taken from the user-provided 67-page lecture note. Explanatory expansions, teaching notes and redrawn schematic diagrams in this edition are original educational elaborations. For external technical context, see the official sources below.

NREL renewable technology overview: https://www.nrel.gov/docs/fy20osti/76198.pdf

U.S. EPA renewable energy technology basics: https://www.epa.gov/re-powering/appendix-b-renewable-energy-technology-basics

NREL 2025 Electricity Annual Technology Baseline: https://atb.nlr.gov/electricity/2025/technologies

NREL System Advisor Model: https://sam.nlr.gov/

Academic integrity and verification. This is an expanded teaching draft, not an institutional endorsement. Students should cite the underlying technical sources in submitted work, verify local standards and current policy data, and avoid treating simplified calculations as final equipment designs.

END OF COMPREHENSIVE EDITION